

MATHEMATICAL READER

Fourth-order vacuum sources and sixth-order energy in cubic SU(2) lattice Yang-Mills theory

A consolidated reader for two exact finite-lattice calculations

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17 September 2026

An introduction to two exact calculations in the same original finite cubic lattice coordinates, followed by all six supplied proof texts. The fourth source and sixth energy are common to both packets. Their further analytic and computational conclusions are distinguished explicitly.

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1 The mathematical problem

This reader brings together two calculations of the fourth coefficient of the logarithmic vacuum and the sixth coefficient of the ground energy for the same finite cubic SU(2) lattice Hamiltonian. Both calculations retain the original links, Haar measure, coupling constants, and open boundary. Their common coefficients are expressed in different coordinate systems. One calculation also supplies bounds on the complete interacting vacuum and physical spectrum, uniform in the size of the finite box. The other supplies explicit signed derivatives of the vacuum coefficients and an independently proved, volume-dependent analytic remainder for the energy.

The purpose of this introduction is to identify the mathematical objects, state the shared results, and explain precisely what each calculation adds. The six proof texts follow in full as Appendices A-F. The introduction is an expository guide to those proofs, not an additional independent analytical verification of them. The associated exact tables and executable evidence are companion material; this volume does not print thousands of polynomial entries. No claim of global historical priority or of a solution of the four-dimensional continuum Yang-Mills problem is made.

1.1 Original lattice and Hamiltonian

Fix an integer $L \geq 2$. The vertices are $\{-L, \dots, L\}^3$; the links are the positively oriented nearest-neighbor edges contained in that box. Let E_L and P_L denote the sets of links and elementary plaquettes, and put $M = |P_L|$. A configuration assigns $U_e \in \text{SU}(2)$ to each positive link. Integration is with respect to product Haar probability on $\text{SU}(2)^{E_L}$. The physical Hilbert space consists of the functions invariant under every vertex gauge transformation, including the transformations at boundary vertices.

For $p = (n; i, j)$, $i < j$, the ordered plaquette holonomy and its fundamental trace are

$$\Omega_p = U_i(n)U_j(n + e_i)U_i(n + e_j)^{-1}U_j(n)^{-1},$$

$$W_p = \text{tr}(\Omega_p), \quad S = \sum_{p \in P_L} W_p.$$

With $T_\alpha = -i\sigma_\alpha/2$ and the original left derivatives $X_{e,\alpha}$, the kinetic operator and Hamiltonian are

$$K = - \sum_{e,\alpha} X_{e,\alpha}^2,$$

$$H = \kappa[K + \xi(2M - S)], \quad \kappa = \frac{2g^2}{a}, \quad \xi = \frac{1}{4g^4}.$$

Here $a > 0$ is the lattice spacing and $g > 0$ the original coupling. In particular, a small value of ξ corresponds to a large value of g . The scalar $2\kappa M\xi$ is part of the original physical energy throughout the calculations.

1.2 The logarithmic vacuum and the coefficient equation

Let $P_H f = \int f dU$ and $Q_H = I - P_H$, and define

$$\Gamma(f, h) = \sum_{e,\alpha} (X_{e,\alpha} f)(X_{e,\alpha} h), \quad B(f, h) = K^{-1} Q_H \Gamma(f, h).$$

The inverse is taken on the nonconstant physical component. At finite coefficient order the papers implement it using the actual Casimir spectrum of the polynomial source. They do not impose a finite-spin approximation to the full interacting Hamiltonian.

The source is the Haar-centered logarithm $v = Q_H \log \psi$ of the positive vacuum. Its equation and formal expansion are

$$v = \xi v_1 + B(v, v), \quad v_1 = \frac{S}{3}, \quad v = \sum_{n \geq 1} \xi^n v_n, \quad v_n = \sum_{i=1}^{n-1} B(v_i, v_{n-i}) \quad (n \geq 2).$$

The source is a function. Its coefficient representatives carry the original multisets of plaquette labels as well; these labels record where each term came from even when an assembled function or channel vanishes. The physical normalization and energy are recovered by

$$c_L = -\frac{1}{2} \log \int e^{2v} dU, \quad \psi_L = e^{v+c_L}, \\ E_{0,L} = \kappa [2M\xi - P_H \Gamma(v, v)].$$

The distinction between this normalized vacuum and auxiliary polynomial representatives is essential. The second calculation also uses the formal eigenline representative $u = 1 + \sum_{\nu \neq 0} u_\nu \lambda^\nu$, with $P_H u_\nu = 0$, for $K - \sum_p \lambda_p W_p$. Its relation to the same logarithmic source is

$$v = Q_H \log u, \quad u = \frac{e^v}{P_H e^v}.$$

The physical diagonal is $\lambda_p = \xi$ for every plaquette; taking it after coefficient calculations retains every directional response and multiplicity.

1.3 Source identity and reading map

The terms *trace packet* and *quaternion packet* below identify two supplied collections dated 17 September 2026. They do not denote different physical theories. The trace packet evaluates original trace words and builds analytic bounds from their coefficient data. The quaternion packet evaluates polynomials in chord variables obtained by an explicit spanning-tree coordinate map, with the original derivatives and Haar measure transported through that map.

Appendix and supplied proof text	Packet and principal role
A. QUARTIC_SOURCE.md	Trace: complete fourth source and coefficient bounds
B. PHYSICAL_RETURN.md	Trace: all-order vacuum correction and physical spectral estimate
C. SIXTH_ENERGY.md	Trace: sixth energy, cube term, remainder, and plaquette observable
D. FOURTH_ORDER_SOURCE.md	Quaternion: complete fourth source in exact tree and quaternion coordinates
E. SIXTH_ORDER_ENERGY.md	Quaternion: sixth energy and a separate finite-volume remainder
F. SIGNED_TANGENT_CERTIFICATE.md	Quaternion: explicit signed directional coefficients and tangent identities

Equation labels in the supplied texts remain local to their original files. Thus “Appendix A, Q14” and “Appendix D, Q15” refer to two presentations of the same single-face component; an

unqualified Q or E label inside an appendix belongs to that source or its named companion. In the sixth-energy formulas the trace packet uses (P, T, C, B) for paths, common-edge triples, corners, and cubes, whereas the quaternion packet uses (P_3, T_e, T_v, C) . The introduction uses the latter notation to avoid that collision. Both use J for unordered adjacent plaquette pairs.

The pinned predecessor is the cubic calculation at Git revision `91434b6962062bd80439d4cb2cae9d2479264dde` in the [Yang-Mills interacting workbench](#). The source texts identify which inherited definitions and bounds they use. Repository status is time-dependent; historical status statements in the appendices describe their source packets.

The appendix proof texts and the algorithms accompanying the edition preserve their supplied source content. Raw conversational input is omitted from the public edition for privacy. References in the original texts to an upload, a session, or a ZIP describe the original provenance; they do not imply that conversational input or every companion file is embedded in this PDF. Formatting and navigation have been added, while the supplied formulas and proofs have been retained. In particular, plain-text mathematical displays are printed in their original notation so that transcription introduces no new mathematical choices.

2 The shared fourth source

2.1 What is being completed

At the quadratic reference $q_2 = \xi v_1 + \xi^2 v_2$, the residual is

$$\xi v_1 + B(q_2, q_2) - q_2 = \xi^3 v_3 + \xi^4 B(v_2, v_2).$$

The fourth coefficient of the solution is, however,

$$v_4 = 2B(v_1, v_3) + B(v_2, v_2).$$

Both packets calculate this complete signed sum. The mixed term $2B(v_1, v_3)$ is indispensable; computing the degree-four part of the quadratic-reference residual alone would not calculate v_4 .

2.2 Completeness and two representations

A nonzero logarithmic coefficient has an edge-connected plaquette multiset. Beginning with a plaquette containing a fixed anchor link, and adding either a repeated plaquette or an edge-adjacent plaquette, gives respectively

$$4, \quad 46, \quad 612, \quad 8621$$

anchored multisets at orders one through four. At order four, signed permutations of the three axes and translations reduce the catalogue to 78 classes:

Partition	4	3 + 1	2 + 2	2 + 1 + 1	1 + 1 + 1 + 1	Total
Classes	1	2	2	19	54	78

Both packets retain explicit maps from these representatives to all 8,621 anchored multisets. Reconstruction in a finite box includes each contained connected multiset once, with the original inverse-link convention and multiplicities.

The trace catalogue has 743 nonzero rational trace-monomial terms. The quaternion catalogue has 4,044 nonzero rational monomials after the sphere relations are used. These counts concern different presentations of the same coefficient functions, and are not a discrepancy in v_4 . A trace monomial can expand into many quaternion monomials. In the quaternion calculation, the variables are the actual chords of the cluster graph: the four-face lateral tube of a cube has five chord variables, which the catalogue retains.

The equality of the intended coefficients is mathematical: both constructions solve the same Haar-centered coefficient equation in the original variables, and the positive Casimir makes its centered solution unique. This identifies their results once the operator maps and residual identities established in the respective proofs are applied. A separate exact reconciliation for this edition also expands all 743 trace terms into the quaternion coordinates and verifies equality of the complete polynomials in all 78 classes, together with agreement of all 8,621 signed-coordinate transports. The edition's accompanying `VALIDATION.md`, in `consolidation/20260917/`, records that reconciliation and its evidence. The two different term counts alone neither prove nor refute that identification.

For one plaquette, the shared closed expression is

$$v_4^{(p^4)} = \frac{17}{10368}\chi_1(\Omega_p) - \frac{7}{51840}\chi_2(\Omega_p),$$

where $\chi_1 = W_p^2 - 1$ and $\chi_2 = W_p^4 - 3W_p^2 + 1$. This is one component of the full catalogue, not a substitute for its mixed-face coefficients.

2.3 Signed tangent data in the quaternion packet

For a plaquette p and a coupling multiindex ρ , the coefficient of the directional derivative of the logarithmic vacuum is

$$Z_{p,\rho} = (\rho_p + 1)v_{\rho+e_p}.$$

Appendix F computes these signed coefficients through total degree three and verifies the differentiated source equation coefficientwise. Its evidence comprises 78 independently reconstructed logarithmic sources, 282 explicitly expanded degree-three response polynomials, and 2,156 residual identities. Of those identities, 282 are the degree-zero equations $KZ_{p,0} = W_p$ and 1,874 are homogeneous positive-degree equations. Thus every residual is zero after subtracting its prescribed forcing. This clarifies the appendix’s compressed phrase “2,156 exactly zero polynomial tangent residuals.” The integer prefactor is the occurrence multiplicity of the differentiated plaquette. These are finite formal-parameter identities; the packet does not infer from them a bound on the full infinite linearized inverse.

3 The shared sixth energy coefficient

3.1 A common coefficient and complementary derivations

Write the ground eigenvalue of $K - \xi S$ near zero as

$$e(\xi) = e_2 \xi^2 + e_4 \xi^4 + e_6 \xi^6 + \dots, \quad E_{0,L} = \kappa(2M\xi + e(\xi)).$$

A link-center involution makes the nontrivial energy branch even. The first two coefficients are

$$e_2 = -\frac{M}{3}, \quad e_4 = \frac{5M}{216} - \frac{2J}{1053}.$$

The trace packet obtains e_6 from the logarithmic-source equation, eliminating v_5 with its exact equation and retaining sources through v_4 . It checks the result against a separate eigenvector recurrence. The quaternion packet uses Haar-centered eigenline coefficients u_1, u_2, u_3 and proves

$$e_6 = -\langle u_3, K u_3 \rangle_H - e_4 \|u_1\|_H^2 - e_2 \|u_2\|_H^2.$$

The two state-normalization terms in this identity are part of the energy calculation. Neither method drops the additive physical scalar or replaces the physical norm by a Euclidean norm of the displayed coefficients.

3.2 Geometric contributions, including the cube

Let P_3 count three-plaquette paths, T_e triples sharing a common edge, T_v triples at a cube corner, and C elementary cubes. The common result is

$$e_6 = -\frac{289}{77760}M + \frac{22285}{23654592}J - \frac{4909}{118272960}P_3 \\ + \frac{244}{4312035}T_e - \frac{212}{542997}T_v - \frac{83}{1944}C.$$

In the trace packet the coefficient for the labelled multiset $p^4 q^2$ is 22285/47309184. The roles p and q may be exchanged. Summing both roles for each unordered adjacent pair gives the coefficient 22285/23654592 of J above. This factor of two is a difference in counting convention, not a disagreement between the packets.

The last term is a three-dimensional closed-surface contribution. Link-center parity shows that the set of odd-multiplicity plaquettes in a nonzero scalar coefficient must have even boundary. At total order six the only nonempty such set is the boundary of an elementary cube. With the six faces oriented outward, Haar contraction gives

$$\left\langle \prod_{p \in \partial c} w_p \right\rangle_H = \frac{2^8}{2^{12}} = \frac{1}{16}.$$

For an ordering π of the six faces let $A_k(\pi)$ be its first k faces, and let $|\partial A_k|$ count the boundary links. The cube coefficient is

$$-\frac{1}{16} \sum_{\pi \in S_6} \prod_{k=1}^5 \frac{4}{3|\partial A_k(\pi)|} = -\frac{1}{16} \frac{166}{243} = -\frac{83}{1944}.$$

All 720 orders are included. The intermediate denominators are the Casimirs of the retained boundary states. Appendix E also derives the same value by pairing complementary three-face subsets.

3.3 Open-box formula

For $m = 2L \geq 4$, the original boundary is included in the exact counts

$$\begin{aligned} M &= 3m^2(m+1), & J &= 6m(3m^2-1), & C &= m^3, \\ T_e &= 12m^2(m-1), & T_v &= 8m^3, \\ P_3 &= 138m^3 - 126m^2 - 24m + 12. \end{aligned}$$

They give

$$e_6 = -\frac{211396463m^3 + 30959193m^2 + 21845782m + 2336684}{4691494080}.$$

Consequently

$$\frac{E_{0,L}}{\kappa} = 2M\xi - \frac{M}{3}\xi^2 + \left(\frac{5M}{216} - \frac{2J}{1053}\right)\xi^4 + e_6\xi^6 + \text{higher even orders}.$$

The coefficient-density limit is $\lim_{m \rightarrow \infty} e_6/M = -211396463/14074482240$. This statement concerns a coefficient. A limit of the full series requires a uniform remainder or a separate convergence argument, as supplied for a particular observable in the trace packet.

4 What is controlled beyond finite order

4.1 The trace packet: full vacuum and physical spectrum

Appendix A derives bounds for the complete fourth source in three auxiliary coefficient norms: a Casimir-weighted local norm, a total-spin budget $m(v_4)$, and a budget $t(v_4)$ at the actual anchor link. These bounds are

$$\begin{aligned}\|v_4\|_{\text{loc}} &\leq \frac{2296826751679}{30073680}, \\ m(v_4) &\leq \frac{17270702970768271}{341697152160}, \\ t(v_4) &\leq \frac{110695177857394584026401}{18025447358750832000}.\end{aligned}$$

These are bounds on coefficient representations used to construct and estimate the vacuum; the physical state norm remains the original weighted Haar norm.

The strongest construction in Appendix B uses the full fourth-order reference $q_4 = \sum_{n=1}^4 \xi^n v_n$ and its entire residual. A convergent Neumann inverse and a convergent nonlinear series construct $v - q_4$, including both infinite tails. The resulting positive function is identified with the actual normalized physical ground state. Its source estimate is then returned to the complete physical excitation problem.

Let Δ_L be the lowest positive excitation energy above that ground state. The result in Appendix B, R30 is

$$\Delta_L \geq \kappa d_{[4]}(\xi), \quad L \geq 2, \quad a > 0, \quad 0 < \xi \leq \alpha_{[4]}.$$

Here $d_{[4]}$ is the explicit expression in R29, and $\alpha_{[4]}$ is the first positive root of the rational polynomial in R25. The stated enclosure is

$$0.018104972231644127075 < \alpha_{[4]} < 0.018104972231644127076.$$

The bound has $d_{[4]}(\xi) > 3/2$ on that whole closed positive interval. In the original coupling, a sufficient rounded condition is $g^2 \geq 3.715960362535435237$. A concrete consequence is

$$\Delta_L > 1.6584 \kappa = 3.3168 g^2/a, \quad g^2 \geq 15/4.$$

This statement is uniform in the finite box size and concerns every physical excitation. Appendix B also preserves an earlier, weaker construction based on q_3 ; its endpoint α_4 must not be confused with the stronger q_4 endpoint $\alpha_{[4]}$ used here.

4.2 Two energy remainders with different domains

The trace packet gives the physical energy remainder after order six as

$$|R_{\geq 8}| \leq \frac{49\kappa|E_L|}{1200} \frac{(60|\xi|)^8}{1 - (60|\xi|)^2}, \quad 0 < |\xi| < \frac{1}{60}.$$

Its analytic input is the inherited uniform logarithmic-source construction on the complex circle $|\xi| = 1/60$. After division by M , the geometric factor $|E_L|/M = (m+1)/m$ is uniformly bounded. The $1/60$ domain of this Cauchy remainder is stated separately from the larger real-coupling interval of the fourth-reference spectral result.

The quaternion packet supplies a separate proof using bounded perturbation of the isolated free vacuum. With

$$R = \frac{3}{8M},$$

its dimensionless energy remainder satisfies

$$|R_8(\xi)| \leq \frac{3}{4} \frac{(|\xi|/R)^8}{1 - (|\xi|/R)^2}, \quad |\xi| < R.$$

The physical remainder is κR_8 . This radius depends on the volume; for $L = 2$ it is $1/640$. The second estimate is independently derived within its packet and does not replace the uniform analytic argument of the first.

4.3 The plaquette observable and the continuum boundary

For the trace packet, differentiating the normalized ground-state eigenvalue yields the mean half-trace

$$P_L(\xi) = \frac{1}{2M} \sum_p \langle W_p \rangle_{\rho_L}, \quad \rho_L = \psi_L^2,$$

and a uniform remainder permits the spatial-volume limit at fixed lattice spacing. Appendix C, E12 gives

$$P_\infty(\xi) = \frac{\xi}{3} - \frac{11}{468} \xi^3 + \frac{211396463}{4691494080} \xi^5 + R_{P,\infty},$$

with

$$|R_{P,\infty}| \leq \frac{49}{40} \frac{(60\xi)^7 [8 - 6(60\xi)^2]}{[1 - (60\xi)^2]^2}, \quad 0 < \xi < 1/60.$$

This is a fixed-lattice-spacing observable. Along the separately specified path $a_n = a_0 2^{-n}$ and $g_n^{-2} = g_0^{-2} + \beta n \log 2$, the spectral domain is $g_n^{-2} \leq 2\sqrt{\alpha_{[4]}}$. For $\beta > 0$ the path eventually leaves that domain. The finite-regulator estimates therefore do not supply a continuum field or a finite positive continuum mass.

5 Literature and evidentiary scope

The exponential-vacuum and gauge-invariant character/Casimir framework has established antecedents. The supplied trace packet explicitly transports the coupling, energy, derivative, and observable conventions of its comparison sources. The quaternion packet checks the one-plaquette coefficient against the Mathieu characteristic value in NIST DLMF. These comparisons support attribution and stated convention checks; they do not provide an independent published verification of the full fourth catalogue or sixth open-box coefficient.

The claims in the appendices combine written arguments with exact finite computation. A finite evaluation at rational link assignments is distinguished there from a polynomial identity, and a coefficient calculation is distinguished from a convergence or spectral theorem. Neither packet claims independent external analytical review or a new Lean certificate. This reader preserves those distinctions.

5.1 References

The identities and versions below were checked against primary records on 17 September 2026. This bibliographic check does not constitute a new audit of the mathematical implications drawn from them; the scope of each use is stated in the supplied proofs.

1. D. Schütte, Zheng Weihong, and C. J. Hamer, *The Coupled Cluster Method in Hamiltonian Lattice Field Theory*. [arXiv:hep-lat/9603026v1](#), 30 March 1996; *Physical Review D* **55** (1997), 2974-2986, [doi:10.1103/PhysRevD.55.2974](#). Exponential-vacuum and character/Casimir framework; convention map in Appendix B, R23.
2. C. H. Llewellyn Smith and N. J. Watson, *The Shifted Coupled Cluster Method: A New Approach to Hamiltonian Lattice Gauge Theories*. [arXiv:hep-lat/9212025v1](#), 18 December 1992; *Physics Letters B* **302** (1993), 463-471, [doi:10.1016/0370-2693\(93\)90428-K](#). Shifted coupled-cluster framework; scope discussed in Appendix C, E7.
3. Thomas Spriggs, Eliska Greplova, Juan Carrasquilla, and Jannes Nys, *Accurate ground states of $SU(2)$ lattice gauge theory in $2+1D$ and $3+1D$* . [arXiv:2509.12323v1](#), 15 September 2025. Explicit coupling, energy, observable, and boundary comparison in Appendix C, E13-E14; the cited version is v1.
4. NIST Digital Library of Mathematical Functions, Chapter 28, *Mathieu Functions and Hill's Equation* (chapter author: Gerhard Wolf), §28.6, *Expansions for Small q* , [equation \(28.6.5\)](#). Release 1.2.8 of 15 September 2026; [official citation information](#). One-plaquette Mathieu coefficient comparison in Appendix E, E17.
5. Timo Jakobs, Marco Garofalo, Tobias Hartung, Karl Jansen, Johann Ostmeyer, Simone Romiti, and Carsten Urbach, *Dynamics in hamiltonian lattice gauge theory: approaching the continuum limit with partitionings of $SU(2)$* . *The European Physical Journal C* **85** (2025), article 1418, published 13 December 2025, [doi:10.1140/epjc/s10052-025-15120-x](#). Single-plaquette Mathieu representation cited in Appendix E, E17.

A The complete connected fourth source in the original cubic SU(2) Hamiltonian

17 September 2026. Parent: 91434b6962062bd80439d4cb2cae9d2479264dde, the unmerged PR8 contribution. This file, PHYSICAL_RETURN.md, and SIXTH_ENERGY.md form one continuation. The exact original-coordinate tables are generated by `verify.py`; every entry and every original anchored multiset is retained. This is a written proof with finite exact computation, not a Lean certificate or an independent external analytical review. No claim of historical priority is made.

A.1 Q1. Original operator and labelled coefficient source

Vertices are $\{-L, \dots, L\}^3$, $L \geq 2$. Edges are every contained positively oriented nearest-neighbor link $e=(n,i)$. Faces $p=(n,i,j)$, $i < j$, carry the original word

$$W_p = \text{tr}(U_i(n) U_j(n+e_i) U_i(n+e_j)^{-1} U_j(n)^{-1}).$$

With $T_\alpha = -i \sigma_\alpha / 2$ and $X_{e,\alpha}$ the original left derivative, retain

$$\begin{aligned} K &= -\sum_{(e,\alpha)} X_{e,\alpha}^2, \\ H &= \kappa K + \kappa \xi^{2M-S}, \\ S &= \sum_p W_p, \quad M = |P_L|, \quad \kappa = 2g^2/a, \quad \xi = 1/(4g^4). \end{aligned} \quad (Q1)$$

The scalar $2\kappa \xi M$ remains. The Hilbert measure is the original product Haar probability. All vertex gauge transformations, including boundary vertices, are imposed in the physical space. The physical H^2 and H^1 domains and positive ground-state identity are as in the pinned original `finite_box_weak_coupling_physical_gap.md`, blob `dfee77c0ca0d6ed897c4172fcbfe23cd9d10d313`.

Let P_H be original Haar integration and $Q_H = I - P_H$. On nonconstant physical Fourier blocks define

$$\begin{aligned} \Gamma(f,h) &= \sum_{(e,\alpha)} (X_{e,\alpha} f)(X_{e,\alpha} h), \\ B(f,h) &= K^{-1} Q_H \Gamma(f,h), \\ v_1 &= S/3, \quad v_n = \sum_{i=1}^{n-1} B(v_i, v_{n-i}). \end{aligned} \quad (Q2)$$

Every occurrence keeps its multiset of original face labels. The support is their original union; cancellation of a coefficient does not delete that support record. At the function level, assembly is the map

$$\begin{aligned} A: \text{direct_sum_multisets } V_{\text{multiset}} &\rightarrow \text{functions}, \\ (f_{\text{multiset}}) &\rightarrow \text{sum } f_{\text{multiset}}. \end{aligned} \quad (Q3)$$

Its kernel is the set of coefficient families with zero assembled Fourier matrix at every original spin label. Neither a zero matrix nor a trace identity is assigned an inverse outside the stated quotient. Scalars removed by Q_H are returned to the ground energy in R9 of the companion file.

A.2 Q2. A finite trace presentation with an explicit evaluation map

An oriented edge word is a tuple $(s_1 e_1, \dots, s_r e_r)$, $s_i = +1$ or -1 . Its evaluation is $\text{tr}(\text{product}_i U_{e_i}^{\hat{s}_i})$, in that order. A monomial is a product of these original trace evaluations. The coefficient algebra is rational. The retained map ev sends a formal rational sum of such monomials to its actual smooth cylinder function.

The program chooses representatives only using the following proved relations in $\ker(\text{ev})$:

- adjacent $U_e U_e^{\hat{(-1)}} = I$, with $\text{tr}(I) = 2$ kept as the scalar 2;
- cyclic trace rotation;
- reversal of the entire word with every sign reversed, since $\text{tr}(V^{\hat{(-1)}}) = \text{tr}(V)$ for V in $\text{SU}(2)$;
- $\text{tr}(V^{\hat{r}}) = P_r(\text{tr } V)$, where $P_0 = 2$, $P_1 = x$, $P_r = x P_{(r-1)} - P_{(r-2)}$.

The last identity follows by multiplying $V^{\hat{2}} - \text{tr}(V)V + I = 0$ by $V^{\hat{(r-2)}}$ and taking the trace. Thus ev factors through the displayed trace relations. It is this factorization, with the original multiset still attached, that relates the coefficient representatives. No faithfulness of an unrestricted trace-word algebra is presumed, and no physical parameter, vector norm, or Haar factor is changed.

Here are the exact derivative formulas used by `trace_algebra.py`. An occurrence of U_e has sign $+1$ and a cut immediately before that occurrence. An occurrence of $U_e^{\hat{(-1)}}$ has sign -1 and a cut immediately after it. If w_i is the cyclic word starting at that cut, then

$$X_{e,\alpha} \text{tr}(w) = \sum_{\text{occ}} s_{\text{occ}} \text{tr}(T_{\alpha} w_{\text{occ}}). \quad (\text{Q4})$$

For two trace factors A, B , the original Pauli matrices give

$$\begin{aligned} \sum_{\alpha} \text{tr}(T_{\alpha} A) \text{tr}(T_{\alpha} B) &= -(1/2) \text{tr}(AB) + (1/4) \text{tr}(A) \text{tr}(B), \\ \sum_{\alpha} \text{tr}(T_{\alpha} A T_{\alpha} B) &= -(1/2) \text{tr}(A) \text{tr}(B) + (1/4) \text{tr}(AB). \end{aligned} \quad (\text{Q5})$$

These follow by substituting $\sum_{\alpha} (T_{\alpha})_{ij} (T_{\alpha})_{kl} = -(1/2) \delta_{il} \delta_{jk} + (1/4) \delta_{ij} \delta_{kl}$.

For one word w and edge e , each occurrence contributes $(3/4) \text{tr}(w)$ to $E_e \text{tr}(w)$. Two different occurrences, with cut segments A, B and signs s, t , contribute

$$st \text{tr}(A) \text{tr}(B) - (st/2) \text{tr}(w). \quad (\text{Q6})$$

For a product of trace factors, retain the entire product rule

$$\begin{aligned} K \text{product } f_i &= \sum_i (Kf_i) \text{product}_{(j \neq i)} f_j \\ &\quad - 2 \sum_{(i < j)} \Gamma(f_i, f_j) \text{product}_{(k \neq i, j)} f_k. \end{aligned} \quad (\text{Q7})$$

Equations Q4-Q7 prove that the rational word operators intertwine evaluation with the original K and Γ , on every original link assignment. They also retain the nonzero cross terms. The separately implemented `differential_audit.py` differentiates quaternion matrix products by explicit generator insertions and never calls these Fierz contractions.

A.3 Q3. Original finite Casimir inverse, with its scalar fibre

For a multiset C of n faces, an edge occurring r_e times has the full spin list

$$j_e = r_e/2, r_e/2-1, \dots, r_e \bmod 2 / 2. \quad (Q8)$$

At each original vertex, a contributing invariant tensor has integral summed spin and $\max j_e \leq \text{sum of the other } j_e$. The necessity follows from the maximum weight of their tensor product. The sufficiency follows by successively adjoining a spin: the attainable spins fill the interval from the excess of the largest spin over the others (or the parity minimum) to their total, in unit steps. This inductively gives spin zero exactly for the two stated tests. For the computation only their necessary direction is needed; retaining extra channels would still give a valid inverse interpolation.

Enumerate all assignments in Q8 satisfying those vertex tests and retain their original Casimirs $c = \sum_e j_e(j_e+1)$. Call the resulting finite set C_C . Every actual representation of the coefficient source lies in that list. Define

$$\begin{aligned} q_C(t) &= \text{product}_{(c \text{ in } C_C, c > 0)} (1-t/c), \\ f_C(t) &= (1-q_C(t))/t - q_C(t) \sum_{(c \text{ in } C_C, c > 0)} 1/c. \end{aligned} \quad (Q9)$$

The quotient by t in Q9 is a polynomial because its numerator has zero constant term. Direct substitution gives $f_C(0)=0$ and $f_C(c)=1/c$ for every positive listed c . The original product-group spectral decomposition therefore proves

$$K f_C(K) f = Q_H f \quad (Q10)$$

on this whole finite coefficient space, with every repeated spin multiplicity retained. This is a polynomial in the original derivatives, not an inverse on an independently chosen matrix. The constant removed by Q10 is recorded explicitly by original Haar integration.

For a face multiset C , let $C=A+B$ range over ordered nonempty proper submultisets, with their original multiplicities. The complete coefficient is

$$\begin{aligned} v_C &= f_C(K) \sum_{(A+B=C)} \text{Gamma}(v_A, v_B), \\ v_{\{p\}} &= W_p/3. \end{aligned} \quad (Q11)$$

For repeated labels each distinct submultiset is counted once. For distinct labels both ordered complementary choices occur. This agrees exactly with Q2 on expansion of the original sums. It proves each computed entry without introducing an eigenfunction cutoff for H : n bounds only the degree of the one Taylor coefficient being calculated.

A.4 Q4. Exhaustive fourth coefficient and inverse coordinate transports

Start at the four original faces incident on the anchor edge $(0,0,0;1)$. Grow a multiset by adjoining a face already present or a face sharing an original edge with a present face. Retain all results as multisets. The cardinalities at orders one through four are

$$4, 46, 612, 8621. \quad (Q12)$$

This exhausts connected anchored multisets: choose a face containing the anchor, take a spanning tree of the adjacency graph of the distinct faces, grow along that tree, and insert each remaining repeated occurrence. This gives a growth order of every such multiset. A disconnected term is zero in Q11 because either an input source is already zero or the two differentiated active supports are disjoint.

The fourth multiset counts and representative counts are respectively

Multiplicity	Original anchored multisets	Cubic representatives
4	4	1
3+1	84	2
2+2	42	2
2+1+1	1572	19
1+1+1+1	6919	54
Total	8621	78

The representative map retains an actual permutation π of the three axes, signs ϵ_i , and an integer translation d :

$$\begin{aligned} y_i &= \epsilon_i x(\pi(i)) - d_i, \\ x(\pi(i)) &= \epsilon_i (y_i + d_i). \end{aligned} \quad (\text{Q13})$$

A positive original edge maps to the corresponding signed edge under Q13. The link-variable map uses that same U or U^{-1} . A face maps to the original face word or its whole inverse, which have equal fundamental traces. Haar integration and each link Casimir are preserved by this map. Applying Q4-Q7 proves that Q13 transports every coefficient in Q11. The full inverse is retained for every one of the 8621 original multisets in `quartic_transports.json`.

No invariance of the auxiliary trace-coefficient norm under separate link inversion is asserted. The estimates in Q6-Q8 below use bounds valid in every original orientation.

`quartic_coefficients.json` contains all 78 polynomials: 743 nonzero rational trace-monomial terms, their original edge dictionaries, face words, multiplicities and complete allowed Casimir lists. The exact self contribution is

$$v_{\{p,p,p,p\}} = (17/10368) \chi_1(\Omega_p) - (7/51840) \chi_2(\Omega_p). \quad (\text{Q14})$$

Here $\chi_1 = W_p^2 - 1$ and $\chi_2 = W_p^4 - 3W_p^2 + 1$. All other contributions are exposed in the table in the same original words. The producer regenerates the entire table; it does not merely test a few displayed examples. Its differential audit checks the original equation for every representative at two declared rational quaternion assignments. Those finite evaluations accompany, rather than replace, the general intertwining proof Q4-Q11.

A.5 Q5. The auxiliary norm and its original physical gauge inequality

For an original nonempty union label S and spin tuple j , write the coefficient as $\text{Tr}(A_{\pi_j}(S, j))$. Define

$$\begin{aligned} ||f||_{\text{loc}} &= \max_a \sum_{(S \text{ contains } a, j \neq 0)} c(j) ||A_{\pi_j}(S, j)||_1, \\ m(f) &= \max_a \sum_{(S \text{ contains } a, j \neq 0)} (\sum_e j_e) ||A_{\pi_j}(S, j)||_1, \\ t(f) &= \max_e \sum_{(S \text{ contains } e, j \neq 0)} j_e ||A_{\pi_j}(S, j)||_1. \end{aligned} \quad (\text{Q15})$$

All these sums concern the explicitly retained coefficient source. The physical norm stays into $\rho |f|^2$ and physical energy stays $\kappa \int \rho \sum |Xf|^2$.

For any active edge e , the endpoint invariant-tensor inequalities force at least $2j_e$ total spin on the other edges at its endpoints. Their remote endpoints are distinct on the triangle-free original cubic graph. Applying the same inequality there and counting every further edge at most twice forces at least j_e further total spin. Thus $\sum_f j_f \geq 4j_e$. Since $j(j+1) \geq 3j/2$ on nonzero half-integers,

$$c(j) \geq 6j_e, \quad m(f) \leq 2\|f\|_{\text{loc}}/3, \quad t(f) \leq \|f\|_{\text{loc}}/6. \quad (\text{Q16})$$

A nonconstant physical block has $c \geq 3$. Products of coefficient matrices have trace-norm bound $\|A \otimes B\|_1 = \|A\|_1 \|B\|_1$. Decomposing the actual tensor representation, pinching to its isotypic blocks and tracing the multiplicity factors gives the complete actual product coefficients with total trace norm at most that product. The last assertion follows by trace duality against block-diagonal contractions; all multiplicities remain. Each original spin generator has operator norm j_e . Summing its three indices and retaining both possible anchored inputs proves

$$\|B(f, h)\|_{\text{loc}} \leq 3[m(f)t(h) + m(h)t(f)] \\ \leq (2/3)\|f\|_{\text{loc}}\|h\|_{\text{loc}}. \quad (\text{Q17})$$

The first source has $(\|v_1\|_{\text{loc}}, m(v_1), t(v_1)) \leq (32, 64/3, 16/3)$. The full second source retains (236, 5834/39, 137/6), from the parent Q2 expansion.

A.6 Q6. Exact cubic channel capacities, including all mixed channels

This stage further evaluates the original cubic table; it changes no coefficient. The per-cluster nuclear masses x are constrained as follows. Every coordinate is nonnegative.

- Pair: $x_0 + x_1 \leq 64, x_0 \leq 16$.
- Repeated cubic: $x_{(1/2)} + x_{(3/2)} \leq 216, x_{(1/2)} \leq 520/3$.
- Path cubic: $x_{00} + x_{01} + x_{10} + x_{11} \leq 512; \quad x_{00} \leq 32; \quad x_{00} + x_{01} \leq 128; \quad x_{00} + x_{10} \leq 128$.
- Common-edge cubic: $x_{(1/2)} + x_{(3/2)} \leq 512; x_{(1/2)} \leq 256$.
- Corner cubic: retain 000, 011, 101, 110, 111, with total at most 512, $x_{000} \leq 8$, and $x_{000} + x_b \leq 128$ for each b in $\{011, 101, 110\}$. Each one-spin channel is exactly zero by the original central-vertex invariant condition.

The pair and path bounds are original single-link Haar contractions. For the repeated cubic, $H = (4/3)W_p P_0(W_p W_q) - (1/3)W_q$ gives $\|H\|_1 \leq (4/3)816 + 8/3 = 520/3$, and the full product $\chi_1(\Omega_p)W_q$ has norm at most $278 = 216$. *For the common-edge cubic, its entire half-isotypic projection is $(2/3)\sum W_r P_0(W_p W_q)$, with norm at most $(2/3)3816 = 256$; this keeps both half-spin copies.* The corner single-zero projections have norm at most $8 \cdot 16 = 128$; its triple-zero projection is one quarter of the original six-edge boundary trace, giving 8. These bounds follow from literal original Haar contractions and Q17, not from independent assignment of channel masses.

A self cubic has the two fixed bounds 8, 64 on its half- and three-half character terms. Maximizing each actual weighted coefficient over these finite polytopes, then summing over the 612 anchored multisets with their actual anchor spins, gives

$$\|v_3\|_{\text{loc}} \leq 918680/351, \\ m(v_3) \leq 336572872/208845, \\ t(v_3) \leq 225985217/1253070. \quad (\text{Q18})$$

The weights are respectively $c(j)|a_j|$, $(\sum j_e)|a_j|$, and $j_{\text{anchor}}|a_j|$ for the complete parent cubic coefficients a_j . `channel_bounds.py` records every rational facet and vertex. The exact vertex calculation solves every candidate set of active facets, retains feasible vertices and evaluates those displayed weights. Since each polytope is bounded, a linear functional has a maximum on that list. This is a finite complete certificate of all three sums. The original v3 coefficients, signs and physical state norm remain unchanged.

A.7 Q7. Three independently valid fourth-source estimates

For each original fourth multiset C, the table bounds its full c-weighted norm by the smallest of the following separately proved scalar upper bounds.

Input-channel bound. For each ordered split $A+B=C$ in Q11, use the actual eigenvectors (a_μ, F_μ) and (a_ν, F_ν) of the lower sources. The c-weighted inverse cancels the output c. Thus the bound is

$$3 \sum_{(\mu, \nu)} |a_\mu a_\nu| (\sum_e j_{\mu, e} j_{\nu, e}) x_\mu x_\nu. \quad (\text{Q19})$$

Maximize it over the two actual mass polytopes of Q6. A bilinear functional on two compact polytopes reaches a maximum at a pair of vertices: fix one variable, maximize the other linearly, then maximize the retained first coordinate linearly. The producer evaluates every such pair, including equal lower multisets without presuming independence of their actual masses. This is a bound; treating their equal masses as independent merely enlarges the admitted domain. Sum every ordered split.

Complete trace expression. Compute Kv_C by Q6-Q7 from the exact table. A nonempty reduced closed word of length l has a tensor trace coefficient of nuclear norm at most 2^{l-1} . To verify it before identifying repeated edge variables, use independent fundamental factors for its occurrences. At a maximum and minimum of the original coordinate height along the closed walk, the alternating two-index contractions each have Euclidean norm $\sqrt{2}$; the other contractions are two-dimensional identities. Separate row/column permutations give 2^{l-t} , with at least one maximum $t \geq 1$. Identifying repeated original links is the actual tensor-representation restriction, followed by the trace-norm-contractive isotypic decomposition used in Q17. Thus the bound 2^{l-1} is valid in every orientation, also for repeated original edges. Products of words multiply these bounds. Applying this to every rational term of Kv_C gives a whole-source bound on $\|v_C\|_{\text{loc}}$; constants cause no difficulty because the nonconstant Fourier sum is at most the full sum. No rotational invariance of this auxiliary norm is used.

Joint output-channel bound. For a four-distinct-face multiset with no edge in more than two faces, choose $s_e=0,1$ on each shared edge. Single-occurrence edges have spin $1/2$. Keep every assignment admitted by the original vertex conditions. Let $c_A(s)$ be the original Casimir of a submultiset A in that assignment. For singleton A set $a_A=1/3$; otherwise define

$$\begin{aligned} a_A(s) &= \sum_{(\text{empty} \neq B \text{ proper subset } A)} \\ & \quad [(c_B(s) + c_A(A \setminus B)(s) - c_A(s)) / (2c_A(s))] \\ & \quad a_B(s) a_A(A \setminus B)(s). \end{aligned} \quad (\text{Q20})$$

A zero c_A is assigned zero by the original Q_H . At every shared edge a subset has either zero factors, one fundamental factor with scalar Casimir $3/4$, or the full pair with the total edge Casimir. Hence these lifted subset Casimirs commute with the final edge projectors. Q10 and the product rule prove Q20 on the full product of the original four traces. A forbidden intermediate component can have a formally displayed coefficient, but its projected function is literally zero; it is not assigned a new vector.

For each set Z of shared edges, integrate precisely those original edge variables. The projection $P_{(0,Z)}$ is $\prod_{e \in Z} (I - E_e/2)$ on this four-face product. Its trace expression, produced by Q6-Q7, supplies the orientation-independent norm bound b_Z just proved. Therefore its final channel masses satisfy

$$\sum_{(s \text{ with } s_e=0 \text{ for all } e \text{ in } Z)} x_s \leq b_Z. \quad (\text{Q21})$$

Maximize $\sum_s c(s) |a_C(s)| x_s$ over $x_s \geq 0$ and every constraint Q21. The file `quartic_bounds.json` retains the entire constraint list, all coefficient weights, a rational primal vector and a rational dual vector y . The checker verifies $y \geq 0$, $A^T y =$ the weights and equality of primal and dual objectives. Consequently $b^T y$ is a proved upper bound independent of any numerical optimization. `exact_lp.py` obtains it by exact rational elimination with Bland's rule; all certification inequalities are checked again. Forty-four of the 78 representatives admit this particular commuting-channel calculation; the others retain Q19 and the complete trace bound.

A.8 Q8. Complete summed bound and physical scope

Sum the selected per-cluster bounds over every original anchored multiset using Q13 and the orientation-independent inequalities above. The complete rational sum is

$$\|v_4\|_{\text{loc}} \leq A_4 := 2296826751679/30073680 < 76374. \quad (\text{Q22})$$

The complete table includes the 78 exact source polynomials and their 8621 transports. The bound itself includes every ordered input split, every admitted joint output channel and every original scalar return. It is stronger than the preceding unevaluated estimate $(128/3) \cdot (944984/351) + 799258/39$.

The three quantities in Q18 and the full fourth coefficient Q22 are the inputs used in `PHYSICAL_RETURN.md`. That argument controls every remaining order. The finite fourth-coefficient calculation is never substituted for the full interacting Hamiltonian.

A.9 Q9. The fourth source's total-spin and actual-anchor-spin bounds

The complete fourth coefficient also gives more information than its c -weighted norm. Retain each original coefficient family and the actual list of allowed spins from Q8. For a representative C with a proved whole c -weighted bound b_C , two immediate estimates are

$$\begin{aligned} m_C &\leq b_C \max_{(j \text{ in } C_C, j \neq 0)} (\sum_e j_e)/c(j), \\ t_{C,e} &\leq b_C \max_{(j \text{ in } C_C, j \neq 0)} j_e/c(j). \end{aligned} \quad (\text{Q23})$$

These follow term by term on the original Fourier coefficients. The maximum is over the complete displayed assignments, not just their distinct Casimir values. It is finite and rational. Every extra admitted spin enlarges these upper bounds.

There is also an estimate directly from the complete v_C trace polynomial. For a monomial m let $r_e(m)$ be its full occurrence count on edge e . Tensor decomposition places every actual output spin at most $r_e(m)/2$. With $b_m = 2^{\sum \text{words}(\text{length}(\text{word})-1)}$, the orientation-independent bound from Q7 gives

$$\begin{aligned} m_C &\leq \sum_m |a_m| b_m (\sum_e r_e(m))/2, \\ t_{C,e} &\leq \sum_m |a_m| b_m r_e(m)/2. \end{aligned} \quad (\text{Q24})$$

Every term and its original occurrences remain. These are bounds on the same canonical Fourier coefficient quantities as Q23; taking the smaller of the two scalar upper bounds does not change a coefficient or physical metric.

For the 44 commuting-channel cases in Q20-Q21, retain the identical mass constraints $A \cdot x \leq b$ and $x \geq 0$. Use instead the two complete objective rows

$$\begin{aligned} \text{weight}_m(s) &= (\sum_e j_e(s)) |a_C(s)|, \\ \text{weight}_e(s) &= j_e(s) |a_C(s)|. \end{aligned} \quad (\text{Q25})$$

Exact rational dual vectors $y \geq 0$ with $A^T y \geq \text{weight}$ bound these quantities by $b^T y$. The producer computes and verifies the primal and dual equations as for Q21. The table `quartic_spin_budgets.json` keeps every dual, the selected scalar bounds and the original edge multiplicities.

For the point-spin sum the anchor must be transported as well. In each actual map Q13 its endpoints 0 and e_1 go to $-d$ and $(\epsilon_i \delta_{(i,1)} - d_i)_i$. The code locates that exact original edge in the representative dictionary and adds its own $t_{C,e}$ bound. The total-spin bound is added once for every original anchored multiset. Thus every original orientation and anchor are retained. In particular this computation uses universal orientation bounds, not a claimed nuclear-norm invariance under a partial transpose.

The resulting complete rational sums are

$$\begin{aligned} m(v_4) &\leq m_4 := 17270702970768271/341697152160, \\ t(v_4) &\leq t_4 := 110695177857394584026401/18025447358750832000. \end{aligned} \quad (\text{Q26})$$

Their approximate sizes are 50543.88912 and 6141.050242. The exact fractions in Q26, rather than those renderings, are used in every subsequent bound. The point-spin result is strictly smaller than $A_4/6$. These are proved bounds for the entire original fourth source, not selected channels or a sampled cluster. `PHYSICAL_RETURN.md` R24-R34 uses them to complete the full correction at q4; R1-R23 retain the earlier cubic-reference calculation as an intermediate result.

A.10 Sources and attribution

The pinned parent provides the complete second and third coordinate calculations, original support reconstruction, and auxiliary coefficient norms; its cubic producer was replayed unchanged. Classical exponential-vacuum and gauge-invariant character/Casimir approaches are described by Schuette, Zheng Weihong and Hamer, arXiv:hep-lat/9603026v1, especially equations 17, 21-26, 30, 38; the actual physical parameter/energy dictionary is retained in the companion return file. No new general coupled-cluster method or worldwide-priority claim is asserted. Source scopes and the latest peer README discovery are in `SOURCE_INTAKE.json`.

B Full corrections at the cubic and fourth sources in the original physical energy

17 September 2026. All operators, parameters, coefficient spaces and source maps are those of QUARTIC_SOURCE.md (Q1-Q26). R24-R34 give the strongest result, using the complete fourth source; R1-R23 preserve the cubic-reference calculation and its explicit maps. This argument constructs the full correction rather than truncating the Hamiltonian. The final estimate is for the original finite-regulator physical spectrum, uniformly in L . The smooth four-dimensional continuum mass is not determined by this contribution.

B.1 R1. The actual cubic reference and its whole residual

Write $x=x_i$ without changing its physical value, and set

$$\begin{aligned} q_3 &= x \cdot v_1 + x^2 \cdot v_2 + x^3 \cdot v_3, \\ R_3 &= x \cdot v_1 + B(q_3, q_3) - q_3 \\ &= x^4 \cdot v_4 + 2x^5 \cdot B(v_2, v_3) + x^6 \cdot B(v_3, v_3), \\ J_3 \cdot h &= 2B(q_3, h), \quad L_3 = I - J_3. \end{aligned} \tag{R1}$$

Every term is an original function. Q_{11} evaluates all of v_4 . The fifth and sixth terms remain as the displayed complete bilinear functions, with the independent bounds below. The Frechet factor two follows by expansion of $B(q_3+h, q_3+h)$ using symmetry of the original Gamma.

Put

$$\begin{aligned} A_4 &= 2296826751679/30073680, \\ m_3 &= 336572872/208845, \quad t_3 = 225985217/1253070, \\ m_2 &= 5834/39, \quad t_2 = 137/6, \\ l_3 &= m_3 + 4t_3 = 292337810/125307, \\ b_{23} &= 3(m_2 \cdot t_3 + m_3 \cdot t_2) = 222621900791/1163565, \\ b_{33} &= 6m_3 \cdot t_3 = 76060493515233224/43616234025. \end{aligned} \tag{R2}$$

The exact Q_{17} source estimate and the three evaluated lower coefficients give

$$\begin{aligned} ||J_3|| &\leq e_{11}(x) = (128/3)x + (3132/13)x^2 + l_3 x^3, \\ ||R_3||_{\text{loc}} &\leq \delta(x) = A_4 x^4 + 2b_{23} x^5 + b_{33} x^6. \end{aligned} \tag{R3}$$

For the cubic part of J_3 , Q_{17} and Q_{16} give $2 \cdot 3[m_3/6 + (2/3)t_3] = m_3 + 4t_3$; this retains the two different spin budgets. The fifth residual coefficient is twice b_{23} , not b_{23} . All sixth-order residual terms are included in b_{33} .

Define the explicitly specified rational polynomial

$$D_4(x) = (1 - e_{11}(x))^2 - (8/3)\delta(x). \tag{R4}$$

Let α_4 be its unique root between $178/10000$ and $179/10000$. Exact rational evaluation proves opposite signs at these endpoints and $e_{11}(179/10000) < 1$. On $[0, 179/10000]$, e_{11} and δ are increasing, and

$$D_4'(x) = -2(1 - e_{11}(x)) \cdot e_{11}'(x) - (8/3)\delta'(x) < 0.$$

Thus α_4 is also the first positive root; $D_4 \geq 0$ and $e_{11} < 1$ on $[0, \alpha_4]$. The full constants in R2-R4 define the polynomial without rounded coefficients.

B.2 R2. Both infinite constructions and their actual residuals

The actual source operator $L3$ has the norm-convergent inverse

$$L3^{(-1)} = \sum_{j \geq 0} J3^j, \quad ||L3^{(-1)}|| \leq 1/(1-\text{ell}(x)). \quad (\text{R5})$$

Let $z = L3^{(-1)}R3$ and $C = L3^{(-1)}B$. Define $w_1 = z$ and $w_n = \sum_{i=1}^{n-1} C(w_i, w_{n-i})$. From R3 and Q17,

$$\begin{aligned} ||w_n||_{\text{loc}} &\leq \text{Cat}_{n-1} c_*^{n-1} z_*^n, \\ z_* &= \delta/(1-\text{ell}), \quad c_* = (2/3)/(1-\text{ell}), \\ \text{theta} &= 4 c_* z_* \leq 1. \end{aligned}$$

The complete series $w = \sum_n w_n$ is absolutely convergent, including $x = \alpha^4$, and satisfies $L3w = R3 + B(w, w)$. Its bound is

$$||w||_{\text{loc}} \leq w_*(x) = (3/4)(1-\text{ell}(x) - \sqrt{D4(x)}). \quad (\text{R6})$$

At $\text{theta} = 1$ the convergence comes from the Catalan series itself. With $b_N = \text{binom}(2N, N)/4^N$, the exact identity $\text{Cat}_N/4^N = 2(b_N - b_{N+1})$ gives the full nonlinear tail

$$\begin{aligned} ||\sum_{n > N} w_n||_{\text{loc}} &\leq (3/4)(1-\text{ell})\text{theta}^{N+1}b_N \\ &\leq (3/4)(1-\text{ell})\text{theta}^{N+1}/\sqrt{N+1}. \end{aligned} \quad (\text{R7})$$

The final inequality follows from $b_N \leq 1/(N+1)$, proved by induction from the displayed ratio $b_{N+1}/b_N = (2N+1)/(2N+2)$. Independently, the finite linear sum has the exact original residual

$$\begin{aligned} L3^{(-1)}R3 - \sum_{j=0}^N J3^j R3 \\ = J3^{(N+1)}L3^{(-1)}R3. \end{aligned} \quad (\text{R8})$$

Its norm is at most $\text{ell}^{N+1}\delta/(1-\text{ell})$. Each $J3$ operation adjoins at most three original faces to the union label, and $R3$ has at most six; the finite linear sums therefore retain supports of at most $6+3N$ faces. R7 retains every subsequent nonlinear union.

B.3 R3. Identification with the original positive vacuum and its scalar

At each finite L the labelled normed coefficient source is complete, and its global c -weighted sum is at most $|E_L|$ times its local norm. The original spin-generator bounds control every first and second derivative by that global sum. Thus the convergent $v = q3 + w$ is C^2 on the original compact product group. The source equation can be summed with its full derivatives. Retain

$$\begin{aligned} C(v) &= \text{int}_H \Gamma(v, v), \\ c_L &= -(1/2) \log \text{int}_H \exp(2v), \\ \psi_L &= \exp(v + c_L), \\ E0, L &= 2\kappa \times M\text{-}\kappa C(v). \end{aligned} \quad (\text{R9})$$

The original product rule gives $H \psi_L = E0, L \psi_L$. Elliptic regularity makes ψ_L smooth and strictly positive. Multiplication and division by ψ_L preserve the original finite-volume H^1 domain. Integration by parts gives, for every smooth f ,

$$q_{-}(H-E_0, L)(\psi_L f) = \kappa \int \psi_L^2 \sum_i |X_i f|^2. \quad (R10)$$

This proves E_0, L is the actual ground energy. A second ground state divided by ψ_L has every original derivative zero and is constant. The source construction is gauge invariant, so this is the actual physical vacuum. The argument does not assume an excitation gap.

The approximate exponential has a separate exact identity, which retains its multiplication defect:

$$\begin{aligned} \exp(-q_3) H \exp(q_3) \\ = \kappa [K - 2\Gamma(q_3, \cdot)] \\ + \kappa [2Mx - C(q_3)] I - \kappa M_{-}(K R_3). \end{aligned} \quad (R11)$$

Here $M_{-}(K R_3)$ means multiplication by the full original polynomial $K R_3$. Q1-Q2 and a direct differentiation prove R11. Its scalar $C(q_3)$ and the multiplication operator remain; they are not replaced by the actual E_0, L before R9 is proved.

The same absolute coefficient arguments give analyticity for complex $|x| < \alpha^4$. At every original power the unique coefficient recurrence is Q2. Consequently this solution agrees, by its actual coefficients and the ground-state identification, with the preceding constructions on their common domain. No competing vacuum or comparison measure is substituted.

B.4 R4. Return to the complete physical spectral problem

The original transported operator is

$$A = \psi_L^{(-1)}(H - E_0, L) \psi_L = \kappa [K - 2\Gamma(v, \cdot)]. \quad (R12)$$

The complete actual point-spin budget is

$$t(v) \leq (16/3)x + (137/6)x^2 + t_3 x^3 + w_*/6.$$

For a physical Fourier function h , the three original generator indices and $\sum_e j_e \leq 2c(j)/3$ give

$$\begin{aligned} ||2\Gamma(v, h)||_X \leq \chi(x) ||Kh||_X, \\ \chi(x) = 4[(16/3)x + (137/6)x^2 + t_3 x^3 + w_*/6]. \end{aligned} \quad (R13)$$

The norm X is the sum of the original trace norms of Fourier coefficients. The scalar output is included in this bound. Substitution of R6 gives the exact identity

$$\begin{aligned} 3(1 - \chi(x)) &= d_4(x) \\ &= (3/2)(1 + \sqrt{D_4(x)}) + (1136/13)x^2 + p_3 x^3, \\ p_3 &= (3/2)m_3 - 6t_3 = 278874091/208845 > 0. \end{aligned} \quad (R14)$$

In particular $\chi < 1$ throughout the entire closed domain. At α^4 , d_4 exceeds $3/2$.

Let $\lambda > 0$ be an actual physical excitation eigenvalue, and h the zero-Haar-mean part of its eigenfunction after division by the actual vacuum. A constant transported eigenfunction has eigenvalue zero, so h is nonzero. The original eigen-equation gives

$$(K - \lambda / \kappa) h = Q_H 2\Gamma(v, h). \quad (R15)$$

Smoothness on this fixed finite product implies absolute summability of its c -weighted Fourier trace coefficients: applying arbitrarily high powers of the elliptic K gives rapid decay, while the number and dimensions of product-spin representations grow polynomially in c at this fixed L . Cauchy-Schwarz then gives the required trace-norm summability. Hence R13 applies to the actual eigenfunction. Q16 gives $c \geq 3$ on every nonconstant physical block. For $0 < \lambda/\kappa < 3$,

$$\begin{aligned} & \| (K - \lambda/\kappa) h \|_X \\ & \geq (1 - \lambda/(3\kappa)) \| Kh \|_X. \end{aligned}$$

Combining this with R13 proves $\lambda \geq \kappa d_4(x)$. Excitations with $\lambda/\kappa \geq 3$ obey the same bound because $d_4(x) = 3(1 - \chi(x)) \leq 3$. The compact physical spectral resolution returns this to the full original physical form domain:

$$\begin{aligned} \Delta_L & \geq \kappa d_4(x), \\ q_A(f) & \geq \kappa d_4(x) \text{Var}_\rho(f), \\ L & \geq 2, \quad a > 0, \quad 0 < x \leq \alpha^4. \end{aligned} \tag{R16}$$

Thus the result includes states whose original labels depend on L . It is a lower bound on all physical excitations, not a lower bound only on a chosen finite trace family.

B.5 R5. Exact numerical enclosures in original physical units

Rational bisection and integer-square comparisons give

$$\begin{aligned} 0.01781868048538520630 & < \alpha^4 < 0.01781868048538520631, \\ 3.745693472404566975 & < 1/(2\sqrt{\alpha^4}) \\ & < 3.745693472404566976. \end{aligned} \tag{R17}$$

The exact condition on the original coupling is $g^2 \geq 1/(2\sqrt{\alpha^4})$. A sufficient rounded condition is $g^2 \geq 3.745693472404566976$. The parent threshold was about 3.825973052393386; the comparison retains that stronger saved value, not an earlier weaker summary.

At $g^2 = 15/4$, $x = 4/225$, the bound is

$$d_4(4/225) > 1.5794.$$

The same rational bound holds for all $g^2 \geq 15/4$. To verify the required monotonicity, for $0 \leq x \leq 4/225$ use $D_4 \leq 1$, $\ell' \geq 128/3$ and R4 to bound

$$\begin{aligned} d_4'(x) & \leq -\frac{3}{2}(1 - \ell(4/225))\frac{128}{3} \\ & \quad + 2\frac{1136}{13}\frac{4}{225} + 3p_3\frac{4}{225}^2 < 0. \end{aligned}$$

Every number in this final test is rational. Therefore

$$\begin{aligned} \Delta_L & > 1.5794 * \kappa = 3.1588 * g^2/a, \\ g^2 & \geq 15/4. \end{aligned} \tag{R18}$$

At $g^2 = 4$, $x = 1/64$,

$$\begin{aligned} d_4(1/64) & > 1.88578, \\ \|v - q_3\|_{\text{loc}} & < 0.019533267617373. \end{aligned} \tag{R19}$$

The unchanged physical scale is $\kappa = 8/a$ there. These bounds include the complete linear and nonlinear tails in R7-R8. The generated verification record contains exact rational brackets, not only these decimal renderings.

B.6 R6. Retained source kernels, complete operator defect, and physical primitive

For the actual finite windows

$$\begin{aligned} V_N &= \text{span}\{R_3, J_3 R_3, \dots, J_3^N R_3\}, \\ V_N &\xrightarrow{L_3} V \xrightarrow{0} 0, \end{aligned}$$

the inclusion in degree zero and the identity in degree one give the commuting support transitions. R5 supplies the explicit isomorphism

$$\begin{aligned} V_{(N+1)}/V_N &\rightarrow \ker[V/L_3 V_N \rightarrow V/L_3 V_{(N+1)}], \\ [h] &\rightarrow [L_3 h], \quad \text{inverse } [L_3 h] \rightarrow [h]. \end{aligned} \quad (\text{R20})$$

Changing a representative by $L_3 V_N$ changes the inverse by exactly V_N , proving both inverse laws. Subtraction of the actual finite primitive sum leaves $J_3^{(N+1)} R_3$ as the representative. The source label remains attached at a receiving zero. This is the actual Split Zero support-indexed complex and its retained primitive, with the same original coefficient domain throughout.

The map from this source to the physical equation has the exact identities

$$\begin{aligned} K L_3 h &= Q_H[K-2\Gamma(q_3, \cdot)]h, \\ Q_H A h - \kappa K L_3 h &= -2\kappa Q_H \Gamma(w, h). \end{aligned} \quad (\text{R21})$$

The right side has X norm at most $(2\kappa/3)w_* \|Kh\|_X$ by the original spin bounds. R11 separately retains the entire multiplication residual and scalar of the reference exponential. Thus no identity between the linearized source and the full physical operator is asserted without its displayed defect.

On centered original physical functions let $d_X f = (X_i f)_i$, with squared derivative norm $\kappa \int \rho_L \sum_i |X_i f|^2$. The inverse on its actual range is $p_X(d_X f) = f$. Its uniqueness follows from the derivative-zero functions being constants and the centering condition. The original ground-state map $f \rightarrow \psi_L f$ and its inverse $h \rightarrow h/\psi_L$ preserve the specified pairings. The variational characterization and R16 give

$$\|p_X\|^2 = 1/\Delta_L \leq 1/(\kappa d_4(x)). \quad (\text{R22})$$

An actual gauge-invariant conditional-expectation kernel is a centered subspace. Restriction of R16 to its original closed form consequently gives the same lower bound for its physical restricted operator. This returns the stronger inverse estimate to the previously computed zero-shift response without changing its forcing or Gram matrix.

B.7 R7. Continuum scope and literature correspondence

The standing path is $a_n = a_0 2^{(-n)}$, $g_n^2 = 1/c_n$, $c_n = g_0^{(-2)} + \beta \log 2$. Its inclusion in R16 is exactly $c_n \leq 2\sqrt{\alpha_4}$. For $\beta > 0$ the path eventually leaves this proved domain. This finite-regulator, volume-uniform result does not assign a finite continuum mass or a nontrivial ultraviolet field to that path.

The exponential vacuum and linear excitation equations have established antecedents. In Schuette, Zheng Weihong and Hamer, arXiv:hep-lat/9603026v1, the coefficient $x_C = 2/g_C^4$ and physical operator $H_C = g_C^2 [K - x_C S]/(2a)$ have the exact original-coordinate return

$$\begin{aligned} g_C &= 2^{3/4} g, \\ H_{\text{original}} &= \sqrt{2} H_C(a, g_C) + 2\kappa \xi M I. \end{aligned} \quad (\text{R23})$$

Substitution verifies both coefficients and the additive energy. Their Hermitian-generator commutator derivative corresponds to iX here, giving the original signs $K v\text{-}\Gamma(v, v)\text{-}\xi S$. The exact physical Haar/energy pairings and complete source residuals in R1-R22 are kept after this comparison. The general method and existence of higher-order calculations are not claimed as new. No global priority or best-known-threshold statement follows from this scoped literature reading.

B.8 R8. Execute the full correction at the calculated fourth source

The fourth table is now used as an actual reference, rather than left as a remainder estimate. Put

$$\begin{aligned} q_4 &= x \ v_1 + x^2 \ v_2 + x^3 \ v_3 + x^4 \ v_4, \\ J_{[4]} h &= 2B(q_4, h), \quad L_{[4]} = I - J_{[4]}. \end{aligned}$$

Expansion of the unchanged source equation Q2 gives the entire residual, with all ordered contributions:

$$\begin{aligned} R_{[4]} &= x \ v_1 + B(q_4, q_4) - q_4 \\ &= x^5 [2B(v_1, v_4) + 2B(v_2, v_3)] \\ &\quad + x^6 [2B(v_2, v_4) + B(v_3, v_3)] \\ &\quad + 2x^7 B(v_3, v_4) + x^8 B(v_4, v_4). \end{aligned} \quad (\text{R24})$$

Each displayed B is the original function with its full union support. The fifth source is not claimed to have been separately expanded into its final channel table. R24 is an exact expression for every part of the reference residual, including the sixth, seventh and eighth degrees.

Use the exact m_4, t_4 from Q26, together with the earlier m_i, t_i . Define

$$\begin{aligned} l_4 &= m_4 + 4t_4, \\ b_{14} &= 3(m_1 t_4 + m_4 t_1), \quad b_{24} = 3(m_2 t_4 + m_4 t_2), \\ b_{34} &= 3(m_3 t_4 + m_4 t_3), \quad b_{44} = 6m_4 t_4, \\ \text{ell}_{[4]}(x) &= \text{ell}(x) + l_4 x^4, \\ \text{delta}_{[4]}(x) &= (2b_{14} + 2b_{23})x^5 + (2b_{24} + b_{33})x^6 \\ &\quad + 2b_{34}x^7 + b_{44}x^8, \\ D_{[4]}(x) &= (1 - \text{ell}_{[4]}(x))^2 - (8/3)\text{delta}_{[4]}(x). \end{aligned} \quad (\text{R25})$$

All of these constants are rational and positive. The underlying estimates are exactly Q17:

$$\|J_{[4]}\| \leq \text{ell}_{[4]}, \quad \|R_{[4]}\|_{\text{loc}} \leq \text{delta}_{[4]}.$$

In particular the fourth contribution to the derivative is $2*3[m_4/6 + (2/3)t_4] = m_4 + 4t_4$, and the factor two in each unequal reference pair is retained.

Let $\alpha_{[4]}$ be the first positive root of the polynomial $D_{[4]}$. On $[0, 182/10000]$, $\text{ell}_{[4]} < 1$, $\text{ell}_{[4]}$ and $\text{delta}_{[4]}$ are increasing, and

$$D_{[4]}' = -2(1 - \text{ell}_{[4]})\text{ell}_{[4]}' - (8/3)\text{delta}_{[4]}' < 0.$$

Exact signs at 181/10000 and 182/10000 therefore prove existence and uniqueness of this root in that interval, and its first-positive-root status. Integer-square comparisons and rational bisection give

$$\begin{aligned}
0.018104972231644127075 &< \alpha_{[4]} \\
&< 0.018104972231644127076, \\
3.715960362535435236 &< 1/(2\sqrt{\alpha_{[4]}}) \\
&< 3.715960362535435237. \quad (R26)
\end{aligned}$$

For every $0 < x \leq \alpha_{[4]}$, $L_{[4]}$ has its actual norm-convergent Neumann inverse and the complete correction $w^1 = v - q_4$ is constructed by the same explicit binary-tree series as R5-R7, with the R25 data. Thus

$$\begin{aligned}
&||L_{[4]}^{(-1)}|| \leq 1/(1 - e_{11_{[4]}}), \\
&||w^{[4]}||_{\text{loc}} \leq w_{[4]}(x) \\
&= (3/4)(1 - e_{11_{[4]}}(x) - \sqrt{D_{[4]}(x)}). \quad (R27)
\end{aligned}$$

For clarity, set $z = L_{[4]}^{(-1)}R_{[4]}$, $C = L_{[4]}^{(-1)}B$, $w_1 = z$ and $w_n = \sum_{i=1}^{n-1} C(w_i, w_{n-i})$. Its scalar majorants are $z_* = \delta_{[4]}/(1 - e_{11_{[4]}})$ and $c_* = (2/3)/(1 - e_{11_{[4]}})$. The parameter $\theta_{[4]} = 4c_* z_*$ lies in $[0, 1]$. The exact complete tail is bounded by

$$\begin{aligned}
&||\sum_{n>N} w_n||_{\text{loc}} \\
&\leq (3/4)(1 - e_{11_{[4]}}) \theta_{[4]}^{N+1} \\
&\quad \text{binom}(2N, N)/4^N \\
&\leq (3/4)(1 - e_{11_{[4]}}) \theta_{[4]}^{N+1}/\sqrt{N+1}. \quad (R28)
\end{aligned}$$

At $\alpha_{[4]}$ this converges to zero by the central-binomial bound. The inverse truncation separately has residual $J_{[4]}^{(N+1)}L_{[4]}^{(-1)}R_{[4]}$ and norm at most $e_{11_{[4]}}^{N+1}\delta_{[4]}/(1 - e_{11_{[4]}})$. These equations control both infinities without replacing their actual coefficient sources. A finite inverse term has original support in at most $8 + 4N$ faces, and every nonlinear union remains recorded.

The resulting $v = q_4 + w^2$ is C^2 in the original finite-volume coordinates by the same global c-summability proof in R3. Its full equation proves R9-R10 with this v , hence identifies the same actual positive physical vacuum and its scalar. The exact approximate exponential formula is R11 with q_3, R_3 replaced by $q_4, R_{[4]}$; the full multiplication defect $K R_{[4]}$ and the scalar $C(q_4)$ remain.

B.9 R9. Strongest full physical lower bound and retained inverse

The actual point-spin bound is now

$$t(v) \leq t_1 x + t_2 x^2 + t_3 x^3 + t_4 x^4 + w_{[4]}/6.$$

The complete physical excitation argument R12-R16 applies to this actual v . Its returned margin is exactly

$$\begin{aligned}
d_{[4]}(x) &= (3/2)(1 + \sqrt{D_{[4]}(x)}) \\
&\quad + (1136/13)x^2 + p_3 x^3 + p_4 x^4, \\
p_4 &= (3/2)m_4 - 6t_4 \\
&= 32092619324045301088/823531037954625 > 0. \quad (R29)
\end{aligned}$$

¹₄
²₄

Indeed expanding $3[1-4(t_1x+t_2x^2+t_3x^3+t_4x^4+w_4)/6]$ using R27 proves R29 term by term. In particular $d_4 > 3/2$ on the full closed positive-coupling interval. The exact original theorem is

$$\begin{aligned} \Delta_L &\geq \kappa d_4(x), \\ q_A(f) &\geq \kappa d_4(x) \text{Var}_{(\rho_L)}(f), \\ L &\geq 2, \quad a > 0, \quad 0 < x \leq \alpha_4. \end{aligned} \quad (\text{R30})$$

The proof includes every physical form-domain vector through the complete compact spectral resolution as in R15-R16. No preassigned missing excitation gap is used in the source construction. A sufficient rounded original coupling condition is $g^2 \geq 3.715960362535435237$; the exact condition is $g^2 \geq 1/(2\sqrt{\alpha_4})$.

At the concrete original parameters,

$$\begin{aligned} g^2 = 15/4, \quad x = 4/225: \quad d_4(x) &> 1.6584, \\ g^2 = 4, \quad x = 1/64: \quad d_4(x) &> 1.89811. \end{aligned} \quad (\text{R31})$$

The first bound holds throughout $g^2 \geq 15/4$. For $x = 4/225$ the exact rational inequality

$$\begin{aligned} &-(3/2)(1-\text{ell}_4(x))(128/3) \\ &+2(1136/13)x+3p_3x^2+4p_4x^3 < 0 \end{aligned}$$

bounds d_4' from above for $0 < x \leq x_0$, because $\sqrt{D_4} \leq 1$ and $\text{ell}_4' \geq 128/3$. Thus d_4 decreases on that interval. It follows in original physical units that

$$\begin{aligned} \Delta_L &> 1.6584\kappa = 3.3168g^2/a, \\ g^2 &\geq 15/4. \end{aligned} \quad (\text{R32})$$

At $g^2 = 4$ the complete reference error obeys $\|v-q\|_{\text{loc}} < 0.011169768000312$. The generated receipt retains full rational brackets for the correction, discriminant, and original energy factors.

For the actual source windows $V_N = \text{span}\{R_4, J_4 R_4, \dots, J_4^N R_4\}$, the two degree maps and inverse in R20 are now the explicit maps

$$[h] \rightarrow [L_4 h], \quad \text{inverse } [L_4 h] \rightarrow [h].$$

They are well-defined on $V_{(N+1)}/V_N$ and the corresponding transported cohomology kernel because L_4 is injective and its inverse is R27. Both inverse compositions remain literal. The receiving scalar, support and full physical defect are

$$\begin{aligned} K L_4 h &= Q_H[K-2\Gamma(q_4, \cdot)]h, \\ Q_H A h - \kappa K L_4 h &= -2\kappa Q_H \Gamma(w_4, h), \\ \|p_X\|^2 &= 1/\Delta_L \leq 1/(\kappa d_4(x)). \end{aligned} \quad (\text{R33})$$

The defect norm is at most $(2\kappa/3)w_4 * \|Kh\|_X$. The physical derivative norm and ground-state multiplication/division maps are exactly those in R22. Thus the stronger inverse also restricts to the original physical conditional kernel, without changing a forcing vector or its raw Gram matrix.

The standing simultaneous path has the exact domain

$$\begin{aligned} a_n &= a_0 2^{(-n)}, \quad g_n^2 = 1/c_n, \\ c_n &= g_0^{(-2)} + \beta n \log 2 \leq 2\sqrt{\alpha_4}. \end{aligned} \quad (\text{R34})$$

For $\beta > 0$ it eventually leaves the certified interval. The root in R26 is the endpoint of this particular proved majorant, not an assigned singularity of the actual theory. This contribution gives no ultraviolet continuum field or finite positive continuum mass. The complete fourth table and its q4 correction are now finished inputs; the next unevaluated coefficient is the original channel-resolved v5.

C The complete sixth-order ground energy and its original cube contribution

17 September 2026. This calculation uses the actual fourth source evaluated in `QUARTIC_SOURCE.md`, not a finite Hamiltonian approximation. The original scalar energy and all extensive and boundary factors are retained. The coefficient is reproduced by an independent eigenvector recurrence, and the cube contribution is additionally computed by all 720 original face orderings.

C.1 E1. Eliminate the fifth source through its exact equation

Retain the original real Haar pairing and source recurrence. Since

$$C(v) = \text{int}_H \Gamma(v, v), \quad E_0 = 2\kappa \sum M\text{-}\kappa C(v),$$

its sixth energy coefficient is

$$\begin{aligned} e_6 := & [\xi^6] (E_0/\kappa) \\ & = -2 \int \Gamma(v_1, v_5) - 2 \int \Gamma(v_2, v_4) - \int \Gamma(v_3, v_3). \end{aligned}$$

The original fifth equation is $Kv_5 = 2Q_H \Gamma(v_1, v_4) + 2Q_H \Gamma(v_2, v_3)$. Integrating against the zero-Haar-mean v_1 , and using $\int \Gamma(f, h) = \int f Kh$, gives

$$\begin{aligned} e_6 = & -4 \int v_1 \Gamma(v_1, v_4) - 4 \int v_1 \Gamma(v_2, v_3) \\ & - 2 \int \Gamma(v_2, v_4) - \int \Gamma(v_3, v_3). \end{aligned} \quad (E1)$$

This identity includes all fifth-source contributions; none is estimated or discarded. For a specified original multiset of six faces, expand every product in E1 over ordered allocations of that multiset to the displayed sources. A repeated label is allocated by its actual multiplicity, as in Q11. The resulting coefficients depend only on sources through degree four, all evaluated in the adjoining tables.

C.2 E2. Exact original Haar integration

Each word is evaluated in its original SU(2) edge variables. For an edge of even total occurrence r , its full spin-zero projector on that polynomial is

$$P_{e,0} = \text{product}_{(j=1, \dots, r/2)} [I - E_e / (j(j+1))]. \quad (E2)$$

Odd occurrence has zero Haar integral, proved by $U_e \rightarrow -U_e$. Formula E2 follows by its values 1 on Casimir 0 and 0 on each original positive spin Casimir in that edge's complete occurrence range. Its output is independent of U_e . Thus replacing U_e by I after E2 is the actual original-edge integration, not a substitution for an unevaluated integral. Repeat over every edge. Q4-Q7 make each step a rational original trace calculation. The code retains the resulting constants, including $\text{tr}(I) = 2$.

The integration algorithm is independently checked on the fundamental character moments 1, 2, 5 at powers 2, 4, 6. For every sixth-order coefficient below, the same final number is obtained from the separate Rayleigh-Schrodinger recurrence for the original $K\text{-}\xi$ S. Its overall vacuum amplitude remains a separate scalar factor; the energy coefficients are independent of that factor because the original eigen-equation is homogeneous in it. The numerical polynomial entries in

that recurrence are coefficient multipliers, not an assigned physical vacuum norm. The actual amplitude and original unit Haar norm are returned by R9.

Explicitly, with multivariate face couplings, write those coefficient multipliers as u_C , $u_{\text{empty}}=1$, $\text{int } u_C=0$ for C nonempty, and keep the overall scalar amplitude a as a separate factor $a u_C$. The energy coefficients satisfy

$$\begin{aligned} e_C &= -\sum_{(p \text{ distinct in } C)} \text{int } W_p u_{(C-\{p\})}, \\ u_C &= K^{-1} Q_H[\sum_{(p \text{ distinct in } C)} W_p u_{(C-\{p\})} \\ &\quad + \sum_{(\text{empty} \neq A \text{ proper submultiset } C)} e_A u_{(C-A)}]. \end{aligned} \quad (\text{E3})$$

Both equations follow by comparing each original face-coupling monomial in $K(a \sum u_C x^C) - \sum_p x_p W_p(a \sum u_C x^C) = e(x)(a \sum u_C x^C)$. The same scalar a appears on both sides; no energy or coupling is rescaled. E3 and E1 give the same six local coefficients. The first calculation uses the logarithmic-vacuum equation, while E3 uses the original linear eigen-equation.

C.3 E3. Exhaustion of the sixth-order connected supports

For each face p , changing U_e to $-U_e$ on one original edge multiplies a source coefficient by the parity of that edge's total original occurrences. Differentiation and all Casimir inverses commute with this center action. A nonzero Haar scalar therefore requires the set of odd-multiplicity faces to have even boundary over $F2$.

A nonempty finite even-boundary set of elementary cubic faces has at least six faces. Equality is the six-face boundary of one original cube. Here is an explicit finite filling argument. Let c_{12}, c_{13}, c_{23} be its binary face arrays. Set the binary cube array

$$b(n_1, n_2, n_3) = \sum_{(k > n_3)} c_{12}(n_1, n_2, k) \bmod 2.$$

The edge-boundary equations imply that the sum over k of c_{12} is constant in n_1 and n_2 ; finite support makes that constant zero. Thus b is finite also toward negative n_3 . Its upper-minus-lower cube boundary has the required c_{12} array. The remaining face difference is a finite2-cycle with $c_{12}=0$. The boundary equations on directions1 and2 force its c_{13}, c_{23} arrays to be constant in direction3; finite support makes them zero. This proves the finite filling and its exact boundary map.

For a nonempty finite cube array b , each nonempty column along an axis has at least two boundary faces normal to that axis. Therefore its boundary has at least twice the sum of the cardinalities of its three coordinate projections, hence at least six. Equality forces each projection to be a singleton, so b is exactly one cube. This proves the stated six-face classification without deleting nonreduced or repeated face labels.

At total order six, all-even multiplicities are $6, 4+2, 2+2+2$. Their connected distinct-face supports are respectively a single face, an adjacent pair, or a path/common-edge/corner triple. The only all-odd surviving support is one complete cube. Disconnected supports have zero logarithmic energy coefficient by Q11 and E1; equivalently the original operator on disjoint active link supports is the sum of its commuting tensor-factor operators, so their scalar eigen-energy has no mixed coefficient. The original scalar and all variables remain in that tensor comparison.

C.4 E4. Complete local coefficient table

The coefficient for each indicated original multiset is

Original multiset	e_C
$\{p,p,p,p,p,p\}$	-289/77760
$\{p,p,p,p,q,q\}, p \sim q$	22285/47309184
$\{p,p,q,q,r,r\}, \text{distinct path}$	-4909/118272960
Same multiplicities, common-edge triple	244/4312035
Same multiplicities, cube corner	-212/542997
Six distinct faces bounding one cube	-83/1944

The first pair row has the separate original role exchange $p \leftrightarrow q$; a sum over unordered adjacent pairs counts both roles. The complete finite evidence includes both pair embeddings (coplanar and perpendicular), all nine original cubic-orbit embeddings of distinct triples, and the exact coefficients computed from both E1 and E3.

For clarity, the four terms in E1 for the cube are

$$-43/1458, -7/972, -13/2916, -1/648.$$

For the single face they are

$$-17/5832, -1/1458, 17/46656, -7/14580.$$

Their sums give the corresponding rows. `sixth_energy.json` retains all four terms for every row, together with their original face words and multiplicities.

C.5 E5. A second, path-resolved cube proof

Orient the six cube faces outward. Reversing a whole SU(2) loop preserves its trace, so this is the exact original trace product. Each of the twelve edges now occurs once as U and once as U^{-1} . The original integral

$$\int U_{ij} \text{conjugate}(U_{kl}) = \delta_{ik} \delta_{jl} / 2$$

supplies twelve factors $1/2$. Its delta identifications leave one free fundamental color at each of the eight vertices. Hence

$$\int \text{product}_{(\text{six cube faces})} W_p = 2^8 / 2^{12} = 1/16. \quad (\text{E4})$$

In the independent six-distinct-face perturbation coefficient, each proper subset has zero scalar energy coefficient by E3 and the even-boundary argument. In any original ordering, edges occurring twice in an intermediate subset occur in no later face. Integration of those edges projects them onto spin zero. Every edge occurring once carries spin $1/2$. The original intermediate Casimir is consequently exactly $(3/4)|\text{boundary}(\text{subset})|$. These projections commute with later independent edge variables and with the original total K . Therefore the full cube coefficient is

$$-(1/16) \sum_{(\pi \text{ in } S_6)} \text{product}_{(j=1)}^5 \frac{4}{[3 |\text{boundary}(\text{first } j \text{ faces of } \pi)|]}. \quad (\text{E5})$$

No intermediate denominator is replaced by the plaquette denominator³. The complete path count is

Five successive boundary lengths	Multiplicity
(4,8,8,8,4)	48
(4,8,8,6,4)	96
(4,6,8,8,4)	96
(4,6,8,6,4)	192
(4,6,6,6,4)	288

There are 720 paths. Their rational sum before the $1/16$ factor is $166/243$. Equation E5 therefore gives exactly $-83/1944$, matching the complete coefficient recurrence. This retains all original intermediate boundaries, their denominators, and the terminal Haar contraction. It is the first closed-surface scalar term on six distinct original faces.

C.6 E6. All original finite-box factors and the complete higher-order remainder

Let $m=2L$. Denote by M, J, P, T, C, B the numbers of original faces, unordered adjacent pairs, three-face paths, common-edge triples, corners and cubes. Their exact values are

$$\begin{aligned} M &= 3m^2(m+1), \quad J = 6m(3m^2-1), \\ P &= 138m^3-126m^2-24m+12, \\ T &= 12m^2(m-1), \quad C = 8m^3, \quad B = m^3. \end{aligned} \quad (E6)$$

For T sum $\text{binom}(r_e, 3)$ over the original edge degrees r_e in $\{2, 3, 4\}$. There are $3m(m-1)^2$ degree-four edges and $12m(m-1)$ degree-three edges, giving the stated T . Each cube has eight three-face corners. To obtain P , count centered face-adjacency wedges and subtract three for each triangular triple. A face degree is $s_i + s_j + 4r_k - 4$, where s_i, s_j are 3 at one of the two cell-end positions and 4 otherwise, and r_k is 1 at either normal boundary and 2 internally. The direct count is

$$\sum_p \text{binom}(\deg p, 2) = 198m^3 - 162m^2 - 24m + 12.$$

Subtract $3(T+C)$, proving P . These formulas preserve every original open boundary. The verifier independently enumerates the actual boxes $L=2, 3, 4$.

Summing all rows in E4 gives

$$\begin{aligned} e6 = & -289M/77760 + 22285J/23654592 - 4909P/118272960 \\ & + 244T/4312035 - 212C/542997 - 83B/1944 \\ = & -(211396463m^3 + 30959193m^2 + 21845782m + 2336684)/4691494080. \end{aligned} \quad (E7)$$

Thus the original full ground energy is

$$\begin{aligned} E0, L = & 2\kappa M \xi - \kappa M \xi^{2/3} \\ & + \kappa (5M/216 - 2J/1053) \xi^4 + \kappa e6 \xi^6 + R_{ge8}. \end{aligned} \quad (E8)$$

The parent analyticity proof on the complex circle $|\xi|=1/60$ bounds the complete scalar $C(v)$ by $49|E_L|/1200$ there. The new source agrees coefficientwise with that proved construction. The exact center involution makes $E0 - 2\kappa M \xi$ even. Cauchy's estimate and the full geometric tail therefore give

$$\begin{aligned}
|R_{ge8}| &\leq (49\kappa |E_L|/1200) \\
&\quad (60|x_i|)^8/[1-(60|x_i|)^2], \\
0 &< |x_i| < 1/60.
\end{aligned} \tag{E9}$$

The finite coefficient polynomial is accompanied by all higher orders through this bound. Its validity range remains stated; it is not silently extended past the Cauchy circle.

At $L=2$, $e_6 = -3528610133/1172873520$. Along the original spatial-volume limit $m \rightarrow \infty$, the coefficient per face is

$$\lim e_6/M = -211396463/14074482240. \tag{E10}$$

This is a limit of the specified Taylor coefficient with its complete original counts. It does not by itself exchange a spatial or ultraviolet limit with a coupling series beyond the proved domain.

C.7 E7. Literature and scope

Schuette-Zheng-Hamer's coupled-cluster/character method and Llewellyn Smith-Watson's shifted linked-cluster formulation are primary antecedents, arXiv:hep-lat/9603026v1 and hep-lat/9212025v1. The former's operator dictionary to Q1 is R23. The latter explicitly describes its truncation and cluster-shift prescriptions; this calculation keeps the original products and uses the full residual bounds R3-R8 and E9. This establishes the method correspondence and the actual all-order error returned here; it is not a claim that the general expansion method or every displayed coefficient is historically new. The scoped search did not establish an independent published match or global priority for the full spatial sixth coefficient E7.

C.8 E8. Return to an actual measured plaquette observable

For real $0 < x_i < 1/60$ at fixed original κ , the constructed simple vacuum is differentiable in x_i . Its original mass is one, hence $2 \operatorname{Re} \langle \psi, \partial_{x_i} \psi \rangle = 0$. Differentiating $H \psi = E_0 \psi$ and testing with the same ψ therefore proves the exact Hellmann-Feynman identity

$$dE_0/dx_i = \kappa (2M - \sum_p \langle W_p \rangle_{\rho}).$$

No derivative of the vacuum is omitted: its two terms cancel through the displayed mass derivative and the actual eigen-equation. Define the original mean half-trace as an explicit observation map

$$P_L(x_i) = (1/(2M)) \sum_p \langle W_p \rangle_{\rho}.$$

Keeping M and the complete coefficients e_4, e_6 from E8 gives

$$\begin{aligned}
P_L(x_i) &= x_i/3 - (2e_4/M)x_i^3 - (3e_6/M)x_i^5 + R_{P,L}, \\
|R_{P,L}| &\leq (49|E_L|/(40M)) \\
&\quad (60x_i)^7 [8 - 6(60x_i)^2]/[1 - (60x_i)^2]^2.
\end{aligned} \tag{E11}$$

To prove the tail, the same original Cauchy bound used in E9 gives $|e_{-}(2n)| \leq (49|E_L|/1200) 60^{-(2n)}$ for every $n \geq 4$. Differentiate its absolutely convergent series on $|x_i| < 1/60$. The exact scalar sum

$$\sum_{n \geq 4} 2n r^{2n-1} = r^7 (8-6r^2) / (1-r^2)^2$$

follows by differentiating $r^8/(1-r^2)$. Its factor 60 and the factor $1/(2M)$ in the actual observation give precisely E11. The ratio $|E_L|/M = (m+1)/m$ is retained and is at most $5/4$ for $L \geq 2$; consequently the entire error is independent of exterior volume after this explicitly defined spatial average.

For each fixed coefficient, all connected source supports have a finite number of original faces. Their interior translates contribute the same coefficient, and the boundary embeddings have vanishing proportion as $m \rightarrow \infty$. The bounds just used are uniform after division by M . Dominated power-series summation therefore proves the full mean-observable limit on $|\xi| < 1/60$, as well as differentiation of its energy-density series. Its first terms are

$$\begin{aligned} P_{\infty}(\xi) &= \xi/3 - (11/468)\xi^3 \\ &\quad + (211396463/4691494080)\xi^5 + R_{P,\infty}, \\ |R_{P,\infty}| &\leq (49/40) \\ &\quad (60\xi)^7 [8-6(60\xi)^2] / [1-(60\xi)^2]^2. \end{aligned} \quad (E12)$$

This is a fixed-lattice-spacing observable return. No continuum scaling limit is used in the coefficientwise or dominated-series argument.

C.9 E9. Explicit correspondence to a current variational benchmark convention

Spriggs, Greplova, Carrasquilla and Nys, arXiv:2509.12323v1, equation1, use the physical convention

$$H_N(g_N, a) = g_N^2/a [K/2 + (4/g_N^4) \sum_p (1-W_p/2)].$$

Specialize this formula to the same original edge and face sets as Q1. Then the literal coefficient substitution is

$$\begin{aligned} g_N &= 2g, \quad \lambda_N = 4/g_N^4 = \xi, \\ H_N(2g, a) &= H_{\text{original}}(g, a), \\ (a/g_N^2) H_N &= H_{\text{original}} / (2\kappa). \end{aligned} \quad (E13)$$

The generator commutators and $E^2 = -\Delta(S_3)/4$ in their equations1-3 agree with the original $T_\alpha = -i \sigma_\alpha/2$ coordinates through their Hermitian derivative iX . Direct substitution in E13 gives exactly κ and $\kappa \xi$ for the two original coefficients. The trace map in their equation4 is $W_p \rightarrow W_p/2$, precisely the measured observation in E11. All energy and trace factors are exhibited.

Their reported finite numerical lattices have their own boundary choice. To retain the comparison on a periodic extension of our same vertex set, let J be pullback from the old edge variables, with inverse adjoint the Haar integral over the additional wrapping links. For $\text{side } 2L+1 \geq 5$ its actual operator defect is

$$\begin{aligned} H_{\text{periodic}} - J - J H_{\text{open}} \\ = \kappa \xi \sum_{(p \text{ in } P_{\text{periodic}} \setminus P_{\text{open}})} (2 - W_p) J. \end{aligned} \quad (E14)$$

The electric terms on the added links annihilate the pullback, and all old terms agree; expansion proves E14. Thus E13 is a convention correspondence on matched index sets, and E14 keeps

the added interactions needed for the other boundary choice. No numerical data from that paper are presented as an evaluation or independent verification of E7 or E12. The earlier 1985 perturbative source cited in that paper was identified bibliographically but its full text was not retrieved here. The new coefficients and remainder are offered as explicit original-convention benchmark data, with their written derivation and exact replay.

D Completed fourth-order source in the original lattice coordinates

Audit and continuation, 17 September 2026. The requested basis is the supplied `Pasted markdown(6).md`. Its original bytes are preserved in `input/`. The published cubic predecessor was read at Git revision `91434b6962062bd80439d4cb2cae9d2479264dde`; source recovery and its limits are in `sources/`. This note gives a complete, executable fourth-order source catalogue. It makes no new claim about a uniform gap or the four-dimensional continuum.

D.1 1. Operator, source, and the distinction that needs to be retained

Use the original finite open cubic graph and positive edges, generators $T_a = -i \sigma_a/2$, derivatives $x_{e,a}$, and Haar probability. Write

$$K = - \sum_{e,a} X_{e,a}^2, \quad H = \kappa[K + \xi(2M - S)], \quad S = \sum_p W_p, \quad \kappa = 2g^2/a, \quad \xi = 1/(4g^4). \quad (Q1)$$

The original ordered face is $w_{(n;i,j)} = \text{tr}(U_i(n)U_j(n+e_i)U_i(n+e_j)^{-1}U_j(n)^{-1})$. Let P_H be Haar expectation and $Q_H = I - P_H$. The original physical source is the zero-Haar-mean logarithmic vacuum $v = Q_H \log \psi$, with scalar return

$$\psi = \exp(v + c), \quad c = -\frac{1}{2} \log \int e^{2v} dU, \quad E_0 = \kappa[2M\xi - P_H \Gamma(v, v)], \quad \Gamma(f, h) = \sum_{e,a} (X_{e,a} f)(X_{e,a} h). \quad (Q2)$$

The scalar c and the additive original energy have not been discarded. The coefficient equation is

$$v = \xi v_1 + B(v, v), \quad v_1 = S/3, \quad B(f, h) = K^{-1} Q_H \Gamma(f, h). \quad (Q3)$$

The inverse in Q3 is on its actual nonconstant physical polynomial image. On a fixed finite graph its uniqueness follows from the positive Casimir and Haar centering.

Consequently, at $q_2 = x_i * v_1 + x_i^2 * v_2$, the exact residual and full fourth coefficient are different specified expressions:

$$R_2 = \xi^3 v_3 + \xi^4 b_{22}, \quad b_{22} = B(v_2, v_2), \quad \boxed{v_4 = 2B(v_1, v_3) + b_{22}} \quad (Q4)$$

The uploaded L5–L7 give b_{22} . Equation Q4 retains the additional incoming source and its unique $\kappa^{-1} Q_H$ primitive. This package computes their signed sum before any norm bound.

D.2 2. The exact tree-coordinate map and its inverse

For the original edge union E of a connected face cluster, choose the stored spanning tree T , root o , and original oriented chords $c_1, \dots, c_r$, where $r = |E| - |V| + 1$. For a vertex v let $h_v(U)$ be the original ordered product along the tree path from o to v , using inverse links on reverse traversals. Define

$$Z_j = h_{s(c_j)}(U) U_{c_j} h_{t(c_j)}(U)^{-1}, \quad \Phi : SU(2)^E \longrightarrow SU(2)^T \times SU(2)^r, \quad U \longmapsto (U_T, Z). \quad (Q5)$$

Its inverse keeps U_T and sets $U_c = h_s(U_T)^{-1} Z_c h_t(U_T)$. Both compositions are literal identities. Successive left and right Haar translations in the chord variables prove $dU = dU_T dZ$. A

gauge-invariant function pulls back from a function of Z invariant under simultaneous conjugation; its original Haar norm is exactly its product-Haar norm in these variables. The remaining root gauge action is retained, rather than treated as an extra independent observable.

At the slice $U_T=I$, an original chord derivative is $L_-(j,a)$, the original left generator of Z_j . For a tree edge e , let $s_e(c), t_e(c)$ be the zero-one indicators that the original paths to the two chord endpoints cross e . Up to the common path-orientation sign for that one e , its field is

$$V_{e,a} = \sum_c [s_e(c)L_{c,a} - t_e(c)R_{c,a}]. \quad (Q6)$$

The common sign cancels in the original squared generator and in Γ . The reduced kinetic operator is the exact original operator

$$K_T = - \sum_{e \in E, a} V_{e,a}^2. \quad (Q7)$$

To prove Q6–Q7, vary one original tree link as $\exp(tT_a)U_e$, and then apply Q5. Each affected chord becomes $\exp(\epsilon s_e tT_a) Z_c \exp(-\epsilon t_e tT_a)$. Its first and second derivatives are precisely the stated fields. Gauge covariance returns this identity from the tree slice to every original configuration. The checker additionally verifies it by independent second-order quaternion jets through every original edge of all 78 representatives. Those are supplementary coordinate tests; Q5–Q7 are the analytic identification of the operator.

D.3 3. Complete polynomial coordinates with exact relations

Each stored chord has quaternion coordinates $Z=q_0 I-i(q_1 \sigma_1+q_2 \sigma_2+q_3 \sigma_3)$, with $q_0^2+q_1^2+q_2^2+q_3^2=1$. The algebra is the rational polynomial quotient by these original sphere relations, one per chord. In the stored representative monomials each q_0 exponent is zero or one. The relation maps are explicit: $q_0^2 \rightarrow 1-q_1^2-q_2^2-q_3^2$; their inverse interpretation is evaluation on the original sphere. The ideal generators have relatively prime leading monomials in different chord groups, so this representation is unique. No state or energy metric is replaced by a Euclidean coefficient norm.

The generator matrices are the original quaternion multiplications by $(0, e_a/2)$ on the left or right. Thus all their entries are rational halves. The original moments are

$$\int q_0^{d_0} q_1^{d_1} q_2^{d_2} q_3^{d_3} dq = 0 \quad \text{if any } d_i \text{ is odd,}$$

$$\int \prod_i q_i^{d_i} dq = \frac{\prod_i (d_i - 1)!!}{4 \cdot 6 \cdots (2 + \sum_i d_i)} \quad \text{when all } d_i \text{ are even.} \quad (Q8)$$

The empty denominator and $(-1)!!$ are one. Products over distinct chords use independent original Haar factors. Q8 supplies every mean and full polynomial Gram used in the package.

D.4 4. The exact finite inverse and coefficient recurrence

Introduce formal face source coordinates λ_p , with the original physical diagonal map $\lambda_p = x_i$ after all coefficients have been computed. The exact physical operator on that diagonal is Q1. For a multiindex ν , its original edge multiplicity is $m_e(\nu) = \sum_{p \text{ containing } e} \nu_p$. Its allowed edge spins are $j_e = m_e/2, m_e/2 - 1, \dots$ down to zero or one half. Therefore the finite list of possible positive Casimirs on that source is

$$\mathcal{C}_\nu = \left\{ \sum_e j_e(j_e + 1) > 0 : j_e \in \{m_e/2, m_e/2 - 1, \dots\} \right\}. \quad (Q9)$$

Extra allowed values are harmless. The implementation proves that its returned inverse solves the original equation, not just a matrix fit. For a centered forcing $\mathbf{f}$, start $\mathbf{r}_0=\mathbf{f}$, $\mathbf{y}_0=0$, and for each c in $\mathbf{C_nu}$ set

$$y_{j+1} = y_j + r_j/c, \quad r_{j+1} = r_j - K_T r_j/c. \quad (\text{Q10})$$

The exact identity is $\mathbf{K_T} \mathbf{y_j} = \mathbf{f} - \mathbf{r_j}$ at every stage. The final polynomial $\mathbf{r_j}$ is checked to be zero in the original sphere quotient. This gives a unique centered inverse because $\mathbf{K_T}$ is a sum of nonnegative squares and its chord fields alone force a zero-derivative function to be constant. All these operations have finite degree because this is a coefficient calculation at a stated order; no finite-spin approximation to the full Hamiltonian or its spectrum is used.

For explicit arithmetic it is convenient to use a formal eigenline representative $\mathbf{u}(\lambda) = 1 + \sum_{\mathbf{nu} \neq 0} \mathbf{u_nu} \lambda^{\mathbf{nu}}$, $\mathbf{P_H} \mathbf{u_nu} = 0$, of $\mathbf{K} - \sum_{\mathbf{p}} \lambda^{\mathbf{p}} \mathbf{W_p}$, with eigenvalue $\mathbf{e}(\lambda)$. This is an explicitly recorded choice of the scalar coordinate of the eigenline. Its exact return is $\mathbf{v} = \mathbf{Q_H} \log \mathbf{u}$, followed by Q2 for the physical unit vacuum. The inverse coordinate relation is $\mathbf{u} = \mathbf{e}^{-\mathbf{v}} / (\mathbf{P_H} \mathbf{e}^{-\mathbf{v}})$, whose denominator has constant coefficient one. The original physical normalization is not dropped. The recurrences used are

$$\begin{aligned} F_\nu &= \sum_{\mathbf{p}: \nu_{\mathbf{p}} > 0} W_{\mathbf{p}} u_{\nu - \mathbf{e}_{\mathbf{p}}}, & e_\nu &= -P_H F_\nu, \\ K_T u_\nu &= Q_H F_\nu + \sum_{0 < \mu < \nu} e_\mu u_{\nu - \mu}, & P_H u_\nu &= 0. \end{aligned} \quad (\text{Q11})$$

Terms are interpreted coefficientwise, with $\mathbf{mu} \leq \mathbf{nu}$. Writing $1 = \log \mathbf{u}$, differentiation of the formal series gives the complete coefficient relation

$$l_\nu = u_\nu - \frac{1}{|\nu|} \sum_{0 < \mu < \nu} |\mu| l_\mu u_{\nu - \mu}, \quad v_\nu = Q_H l_\nu. \quad (\text{Q12})$$

As an independent polynomial check for each coefficient the program verifies

$$K_T v_\nu = Q_H \sum_{0 < \mu < \nu} \Gamma_T(v_\mu, v_{\nu - \mu}), \quad |\nu| > 1, \quad (\text{Q13})$$

with $\mathbf{K_Tv}(\mathbf{e_p}) = \mathbf{W_p}$ and $2 \mathbf{Gamma_T}(\mathbf{f}, \mathbf{h}) = (\mathbf{K_T} \mathbf{f})\mathbf{h} + \mathbf{f}(\mathbf{K_T} \mathbf{h}) - \mathbf{K_T}(\mathbf{f}\mathbf{h})$. These are identities of complete rational polynomials, not agreements only at finitely many sampled configurations.

D.5 5. All connected fourth-order supports are included

The source operation Gamma differentiates a common original link. Its inverse Casimir creates no new link. Consequently every nonzero logarithmic coefficient has an edge-connected face multiset. Starting from the four faces incident on the original anchor edge $(0,0,0; \text{direction } 0)$, the following exhaustive recursion suffices: add a repeated present face or a face sharing an original edge with a present face, and retain the sorted multiset. Any connected multiset containing that anchor can be recovered in this way by removing a leaf of a rooted spanning tree of its adjacency graph, or one repeated occurrence. Counts by order are

$$4, \quad 46, \quad 612, \quad 8621. \quad (\text{Q14})$$

The 48 signed coordinate permutations and translations act on the original faces and links. A reversed edge maps to its inverse; this is a Haar- and Casimir-preserving coordinate map. A reversed complete SU(2) face word has the same trace. Each orbit is represented by its explicitly stored minimum corner coordinates and ordered axes. The fourth-order orbit counts are

Face-multiplicity partition	Orbits
4	1
3+1	2
2+2	2
2+1+1	19
1+1+1+1	54
Total	78

`results/quartic_coefficients/class_00.json` through `class_77.json` contain all 4,044 nonzero rational monomials of these 78 fully calculated source coefficients. They also give original faces, tree links, chords, root, original derivative fields, coupling multiindex, Haar mean and full Haar state/energy norms. These auxiliary Haar norms are not asserted to be the interacting physical norms.

One class (index25) has four distinct faces but five chord variables: the four-sided tube formed by the cube's lateral faces. Its original graph has an extra independent cycle. The implementation retains it. Replacing the chord count by the number of faces would change this original source.

The source for any original finite box is reconstructed by summing each contained connected multiset once, pulling its representative polynomial back through the stored lattice symmetry and Q5, and setting every formal face coordinate to xi. Q11–13 already include the correct multiplicities; no additional division by 4! is applied. `results/anchored_quartic_transports.json` also records all 8,621 original face multisets, their class numbers, and an explicit signed coordinate permutation plus translation for each. The original edge orientation and inverse-link rule are implemented by `transform_edge` in `calculations/reconstruct.py`. This is a complete finite, coordinate-level specification of v4, rather than an unevaluated B term.

D.6 6. A readable closed component and the signed derivative

For a single face the fully evaluated contribution is

$$v_4^{(p^4)} = \frac{17}{10368} \chi_1(\Omega_p) - \frac{7}{51840} \chi_2(\Omega_p). \quad (\text{Q15})$$

The own-face part of the already available b22 is `chi1/10368-chi2/31104`. The other incoming term in Q4 has coefficients `chi1/648-chi2/9720`. Adding them gives Q15 exactly. The signs remain in the catalogue before any norms are taken.

The catalogue also determines every formal plaquette-direction response through degree three by the exact derivative

$$Z_p^{[3]} = \partial_{\lambda_p} \sum_{1 \leq |\nu| \leq 4} \lambda^\nu v_\nu. \quad (\text{Q16})$$

Differentiating Q13 coefficientwise proves that the terms through total degree three in `[K-2Q_H Gamma(q3,.)] Z_p^[3] - W_p` vanish, with `q3=sum_(1<=|nu|<=3) lambda^nu v_nu`. The remaining degree4–6 polynomial is exactly the corresponding sum of those unremoved higher products. This records the signed tangent data; it is not being promoted to a uniform bound on the full linearized inverse.

The previous numerical coupling threshold requires a coefficient trace-norm bound and a full infinite-tail argument. This session does not replace those with the catalogue's Haar norms or assert a new threshold.

D.7 7. Evidence scope

Every returned coefficient has zero Haar mean, an exact inverse residual, and an independently expanded exact logarithmic-source residual. The sum over all 78 coefficient downsets executes 1,053 inverse/source steps. The additional second-order-jet audit differentiates original unfixed links and verifies the tree return and original vertex-gauge invariance at retained rational assignments for every representative. Those assignments are an independent regression; the polynomial equalities and operator maps above are the identity argument. No independent external or Lean review is claimed. The full raw input, recovered code, calculations and receipts are included in this session's ZIP.

E Completed sixth-order ground-energy calculation, including the cube surface

Audit and continuation, 17 September 2026. This result is derived in this session. The uploaded status line about sixth-order coefficients contained no such coefficient table, proof, or execution record. The formulas below retain the original Hamiltonian and open-box geometry. They are finite-order coefficients with a separately proved finite-volume remainder, not a new uniform-coupling or continuum mass-gap theorem.

E.1 1. Original parameters and exact coefficient convention

Use Q1 of FOURTH_ORDER_SOURCE.md:

$$H = \kappa\{K + \xi(2M - S)\}, \quad S = \sum_p W_p, \quad \kappa = 2g^2/a, \quad \xi = 1/(4g^4). \quad (\text{E1})$$

Write the ground eigenvalue of $K\text{-}\mathbf{x}_i$ S as $\mathbf{e}(\mathbf{x}_i) = \mathbf{e}_2 \mathbf{x}_i^2 + \mathbf{e}_4 \mathbf{x}_i^4 + \mathbf{e}_6 \mathbf{x}_i^6 + \dots$. Then the original physical energy is $E_0 = \kappa(2M \mathbf{x}_i + \mathbf{e}(\mathbf{x}_i))$. The additive original scalar stays in this equation. A link-center transformation $U_{\mathbf{n},i} \rightarrow (-1)^{\sum_{j<i} \mathbf{n}_j} U_{\mathbf{n},i}$ preserves Haar and K and changes every W_p to $-W_p$. It intertwines $K\text{-}\mathbf{x}_i$ S with $K\text{+}\mathbf{x}_i$ S; the ground branch near zero is simple, so $\mathbf{e}$ is even.

Use the formal eigenline representative $\mathbf{u} = 1 + \mathbf{x}_i \mathbf{u}_1 + \mathbf{x}_i^2 \mathbf{u}_2 + \dots$, $P_H \mathbf{u} = 0$. The exact scalar return to the original unit vacuum is Q2 and Q12 of the companion. The low-order equations are

$$\begin{aligned} u_1 &= K^{-1}S = S/3, & u_2 &= K^{-1}Q_H(Su_1), & u_3 &= K^{-1}Q_H(Su_2 + e_2u_1), \\ e_2 &= -\langle S, u_1 \rangle_H = -M/3, & e_4 &= -\langle S, u_3 \rangle_H = 5M/216 - 2J/1053. \end{aligned} \quad (\text{E2})$$

Here J counts original unordered pairs of plaquettes sharing an edge. All pairings are original Haar pairings.

The complete sixth coefficient depends only on these three actual wavefunction coefficients:

$$e_6 = -\langle u_3, Ku_3 \rangle_H - e_4 \|u_1\|_H^2 - e_2 \|u_2\|_H^2. \quad (\text{E3})$$

For an explicit proof, use $\mathbf{e}_6 = -\langle S, \mathbf{u}_5 \rangle = -\langle \mathbf{u}_1, Ku_5 \rangle$. The order-five equation is $Ku_5 = Q_H S \mathbf{u}_4 + \mathbf{e}_2 \mathbf{u}_3 + \mathbf{e}_4 \mathbf{u}_1$. Move K through $\langle S \mathbf{u}_1, \mathbf{u}_4 \rangle$ by $Ku_2 = Su_1 + \mathbf{e}_2$, and then use $Ku_4 = Q_H S \mathbf{u}_3 + \mathbf{e}_2 \mathbf{u}_2 + \mathbf{e}_4$. Finally substitute $Su_2 = Ku_3 - \mathbf{e}_2 \mathbf{u}_1$. The two cross terms $\mathbf{e}_2 \langle \mathbf{u}_1, \mathbf{u}_3 \rangle$ cancel; the remaining terms are precisely E3. This also shows why the two state-denominator corrections in E3 must be retained.

E.2 2. Complete third-vector spectral data

For one face the spin-3/2 character component of $\mathbf{u}_3$ is $\chi_{\mathbf{u}_3}(3/2) = 0$. Its kinetic norm contribution is $1/8640$. The coefficient of the original W_p in the assembled $\mathbf{u}_3$ is

$$a_p = -5/216 + d_p/1053, \quad (\text{E4})$$

where d_p is its original plaquette adjacency degree. This is obtained by multiplying the self and adjacent components of $\mathbf{u}_2$ by each possible outer face in $S \mathbf{u}_2 + \mathbf{e}_2 \mathbf{u}_1$. It is also the coefficient required by $\mathbf{e}_4 = -\sum_p a_p$ and $\sum_p d_p = 2J$.

For the repeated multiset p, p, q on adjacent faces, put $F = (W_p^2 - 1)W_q$. The two actual shared-edge spins are $1/2$ and $3/2$, with Casimirs 9 and 12. The coefficients in $\mathbf{u}_3$ are

$$r_{1/2} = \frac{1/24 + 1/9 + 1/39}{9} = \frac{167}{8424}, \quad r_{3/2} = \frac{1/24 + 4/39}{12} = \frac{5}{416}. \quad (\text{E5})$$

Their Haar squared masses are $1/3$ and $2/3$. To verify the lower mass, write the original shared unit quaternion u and complementary unit quaternions v, w . The harmonic projection is $\mathbb{H} = (8/3)(u \cdot v)(v \cdot w) - (2/3)(u \cdot w)$. Using $\text{int } u_i u_j = \delta_{ij}/4$ and independent Haar in v, w gives $\langle \mathbb{H}^2 \rangle = 4/9 + 1/9 - 2/9 = 1/3$. The full $\langle F^2 \rangle = 1$, giving the other mass. Thus each oriented repeated-adjacent contribution to $\langle u_3, Ku_3 \rangle$ is

$$R_{\text{adj}} = 3r_{1/2}^2 + 8r_{3/2}^2 = 110453/47309184.$$

A repeated nonadjacent pair has coefficient $1/72$, Casimir 11, Haar squared norm one, hence contribution $R_{\text{dis}} = 11/5184$.

For a distinct pair in u_2 the adjacent coefficients are $u_0 = 4/27$, $u_1 = 4/39$, with shared-edge masses $1/4$ and $3/4$. The adjacent-pair squared norm in u_2 is therefore

$$P_2 = \frac{1}{4}(4/27)^2 + \frac{3}{4}(4/39)^2 = 1648/123201. \quad (\text{E6})$$

The nonadjacent-pair value is $1/81$, and the self value is $1/576$.

For three distinct faces the original projection masses and Casimirs are as follows. Each coefficient is the sum of its three possible outer-face sources from E2, divided by that actual Casimir.

- No adjacency: coefficient $1/27$, Casimir 9, squared mass one.
- One adjacent pair: $u_s/3$, Casimir $15/2 + 2s$, masses $1/4, 3/4$.
- Two-edge path: coefficient $(1/9 + u_s + u_t)/(6 + 2s + 2t)$, with masses $(1, 3, 3, 9)/16$ for shared-edge spins $(0, 0), (0, 1), (1, 0), (1, 1)$.
- Common-edge triple: coefficients $(3/2)(u_0 + u_1)/(15/2)$ and $3u_1/(21/2)$; squared masses $1/2, 1/2$.
- Cube corner: coefficient $[(3-n)u_0 + n u_1]/(9/2 + 2n)$; the $n=0, 2, 3$ squared masses, with $n=2$ aggregated, are $1/16, 9/16, 3/8$. The $n=1$ subspace is killed by the original vertex invariant projection.

Here is a derivation of all nontrivial masses. For a path, one shared-edge Haar projection is one half of the original two-face boundary times the third trace, with squared norm $1/4$. Both such projections give one quarter of the eight-link boundary trace, with squared norm $1/16$. Orthogonality of the commuting original edge projections gives the remaining three masses. For a common edge, expose three independent complementary unit quaternions and the shared original unit quaternion. Each pair singlet term has squared norm $1/4$, and each distinct cross pairing is $1/16$. The identity $P_{-}(1/2)F = (2/3)\sum W_r P_{-0}(W_p W_q)$ gives mass $(4/9)(3/4 + 6/16) = 1/2$; its complement has the other half. For a corner, the triple-singlet projection is one quarter of the six-link boundary, of squared mass $1/16$. Each single-edge-singlet projection has squared mass $1/4$. Its only additional component is one of the three $n=2$ channels, of mass $3/16$. The full mass is one, so the $n=3$ remainder is $3/8$. All simple-loop Haar squared norms here equal one by integrating one of its original links. Unique exterior links justify the stated product masses.

These give the following exact kinetic contributions for each distinct triple in u_3 :

Adjacency type	Contribution to $\langle u_3, Ku_3 \rangle$
none	$1/81$
one pair	$4768/369603$
path	$1595629/118272960$
common edge	$20032/1437345$

Adjacency type	Contribution to $\langle u_3, K u_3 \rangle$
cube corner	210880/14660919

The direct original-generator calculation in `gauge_polynomial.py` constructs u_1, u_2, u_3 without this table, using the exact K fields and inverse residuals. It agrees with E3 and the table for every geometric orbit of a connected one-, two-, or three-face set. That calculation also verifies the complete Rayleigh numerator and denominator identity, preserving all cross terms.

E.3 3. Linked contributions and why six distinct cube faces occur

With independent formal face couplings, an energy coefficient with support split into edge-disjoint components is additive in those components. The full Haar space, electric operator and potential factor over the disjoint edge sets; their positive vacuum product is already gauge invariant at any shared vertices. Thus mixed connected-energy coefficients vanish on an edge-disconnected set. Subtracting proper subcluster energies gives the following linked sixth-order weights:

Support	Weight
One plaquette	-289/77760
Adjacent pair, with all sixth-order powers on that support	22285/23654592
Distinct path of three plaquettes	-4909/118272960
Three plaquettes sharing an edge	244/4312035
Three faces at a cube corner	-212/542997
All six faces of an elementary cube, each once	-83/1944

For example, the original two-adjacent-face energy has $e_6 = -767713/118272960$. Subtracting twice the one-face value gives the pair weight. The original path/common/corner three-face e_6 values are $-183461/19712160$, $-5281/638820$, and $-2555051/293218380$ respectively. Subtracting the three one-face values and the actual number of adjacent pair weights gives the table. These are rational identities, executed both from E3 and from the independently constructed polynomial states.

The last row needs an explicit additional calculation. Let Z_e multiply one original link by $-I$. It commutes with K and preserves Haar. The operator

$$\Pi_{\text{even}} = \prod_e (I + Z_e)/2 \quad (\text{E7})$$

projects the original polynomial source onto the edge-even subspace. A monomial in face couplings can contribute to the scalar energy only if its set of odd-multiplicity faces is a mod-two closed cubical surface. For total order six, either every face has even multiplicity (supports of at most three faces), or the six distinct faces form an elementary cube surface.

For completeness, every finite cubical two-cycle in this open cubic box bounds a finite mod-two set of cubes. This can be constructed by assigning cube membership from the exterior along one coordinate: the cycle condition ensures the assignments obtained from the other faces agree. For a nonempty cube set, each occupied coordinate line has two boundary faces. Two distinct cubes give at least five occupied two-dimensional projection cells in the three directions, hence at least ten boundary faces. A boundary with at most six faces therefore comes from exactly one cube. This proves the stated exhaustion at order six.

Exact cube calculation

Orient the six original face words outward. Complete face reversal leaves an SU(2) fundamental trace unchanged. Each of the twelve original edges is then traversed once in each direction. The identity $\int U_{ab} \text{conjugate}(U_{cd}) dU = \delta_{ac} \delta_{bd} / 2$ supplies one factor 1/2 per edge. The remaining trace-index identifications have eight independent vertex-color loops, each of dimension two. Consequently

$$\left\langle \prod_{p \in \partial c} W_p \right\rangle_H = 2^8 / 2^{12} = 1/16. \quad (\text{E8})$$

The coordinate union-find audit retains all twelve edge gluings and verifies the eight color classes.

For an order of the six distinct face insertions, let A_k be its first k faces. Every edge internal to A_k has both its insertions already present. Any nontrivial spin on that edge is killed by final Haar expectation, because no later face contains it. Its singlet projector commutes with the remaining insertions and the Casimir. Each boundary edge carries its original spin 1/2, so the retained intermediate energy is $(3/4) |\text{partial } A_k|$. No proper A_k is a closed surface, so no intermediate vacuum subtraction occurs. Therefore the full linked coefficient is

$$-\frac{1}{16} \sum_{\pi \in S_6} \prod_{k=1}^5 \frac{4}{3|\partial A_k(\pi)|} = -\frac{1}{16} \frac{166}{243} = \boxed{-83/1944}. \quad (\text{E9})$$

All 720 original orders and their five boundary counts are in `results/cube_sixth_paths.json`.

There is also an independent three-plus-three calculation. Of the twenty three-face cube subsets, twelve are paths and eight are corners. Their all-internal-singlet u_3 coefficients are 11/162 and 8/81, with boundary energies 6 and 9/2. The cross pairing with the complementary triple is E8. Thus the contribution to E3 is

$$-\frac{1}{16} [12 \cdot 6(11/162)^2 + 8 \cdot (9/2)(8/81)^2] = -83/1944. \quad (\text{E10})$$

The cube term is a required original three-dimensional contribution. It cannot be assigned a planar value by deleting the extra faces.

E.4 4. The complete finite-box coefficient

Let M count original plaquettes, J adjacent pairs, P_3 three-face paths, T_e common-edge triples, T_v cube-corner triples, and C cubes. The result is

$$e_6 = -\frac{289}{77760}M + \frac{22285}{23654592}J - \frac{4909}{118272960}P_3 + \frac{244}{4312035}T_e - \frac{212}{542997}T_v - \frac{83}{1944}C. \quad (\text{E11})$$

This formula can also be verified before linked subtraction. The full $\|u_2\|^2$ is $M/576 + (c_{\text{hoose}}(M,2) - J)/81 + J*(1648/123201)$. The full $\langle u_3, Ku_3 \rangle$ has: $3 \sum_p a_p^2 + M/8640$; the repeated contributions $[M(M-1) - 2J]*(11/5184) + 2J*(110453/47309184)$; the distinct-triple table in section2; and the positive norm cross term $(83/1944)C$. Inserting these into E3 cancels every disconnected term and gives E11.

For $m=2L \geq 4$, the original open-box counts are

$$\begin{aligned}
M &= 3m^2(m+1), \quad J = 6m(3m^2-1), \quad C = m^3, \\
T_e &= 12m^2(m-1), \quad T_v = 8m^3, \\
P_3 &= 138m^3 - 126m^2 - 24m + 12.
\end{aligned} \tag{E12}$$

Here $T_e = \sum_e \text{choose}(r_e, 3)$ for the original incidence r_e ; each cube has eight different corner triples. For paths, count the wedges centered at an original face, then subtract three per adjacency triangle: $P_3 = \sum_p \text{choose}(d_p, 2) - 3(T_e + T_v)$. In one orientation, faces in each boundary coordinate plane have degrees $8 - b_x - b_y$, and those in an interior plane have $12 - b_x - b_y$, where b_x, b_y indicate whether the face interval touches its corresponding boundary. The numbers for $b_x + b_y = 0, 1, 2$ are $(m-2)^2, 4(m-2), 4$. There are two boundary and $m-1$ interior planes and three orientations. Their exact sum is $198m^3 - 162m^2 - 24m + 12$, giving E12. The executable also enumerates all original faces and triples directly for $L=2, 3, 4$.

Substitution yields the closed finite-box coefficient

$$e_6 = -\frac{211396463m^3 + 30959193m^2 + 21845782m + 2336684}{4691494080}. \tag{E13}$$

In particular, at $L=2$,

$$\frac{E_0}{\kappa} = 480\xi - 80\xi^2 + \frac{1198}{351}\xi^4 - \frac{3528610133}{1172873520}\xi^6 + R_8(\xi). \tag{E14}$$

The limit of this exact coefficient divided by M is $-211396463/14074482240$. This is a limit of a finite-order coefficient; no interchange with an infinite-volume perturbative sum is asserted.

E.5 5. An independently proved finite-volume remainder

The original physical free Casimir has lowest nonconstant eigenvalue 3. A physical nonconstant spin support has no degree-one active vertex; a finite graph of minimum active degree two contains a cycle, and the original cubic graph has no cycle shorter than four. Each active spin has Casimir at least $3/4$. Four spin-one-half edges around one original face attain 3. This proves the free value used here.

On the physical space, $\|S\| \leq 2M$. On the contour $|z|=3/2$, the free resolvent $(K-z)^{-1}$ has norm at most $2/3$. Set

$$R = \frac{3}{8M}. \tag{E15}$$

For $|x| \leq R$, the bounded perturbation $-x S$ has norm at most $3/4$. The contour resolvent Neumann series is thus bounded by a ratio at most $1/2$, and its isolated spectral projection has the same rank one as at zero. This gives an analytic ground branch $e(x)$. The elementary resolvent inclusion $\text{dist}(z, \text{spec } K) > \|x S\| \Rightarrow z \text{ in resolvent}(K - x S)$ puts its single enclosed eigenvalue in $|e(x)| \leq 3/4$; the other free eigenvalues are at least 3 away from zero. On the real interval this branch is the original ground energy after subtracting the explicit scalar $2Mx$.

Cauchy's estimate on the displayed radius and the exact parity in E1 give

$$|R_8(\xi)| \leq \frac{3}{4} \frac{(|\xi|/R)^8}{1 - (|\xi|/R)^2}, \quad |\xi| < R. \tag{E16}$$

The physical remainder is κ times E16. For $L=2$, $R=1/640$. This complete remainder is volume dependent. It uses no unverified uniform logarithmic-source bound and does not extend an earlier coupling threshold or prove a continuum mass gap.

E.6 6. Primary-literature check with explicit conventions

For one original plaquette, K acts on class functions of $\Omega = \cos(\theta)I - i \sin(\theta)\mathbf{n} \cdot \boldsymbol{\sigma}$ as $-(\partial_\theta^2 + 2\cot(\theta)\partial_\theta)$. Put $u(\theta) = \sin(\theta)f(\theta)$ and $\theta = 2z$. The equation for κ - ξ w becomes the Mathieu equation with $a = 4(e+1)$ and $q = -4\xi$, with zero boundary values at $z=0, \pi/2$. Thus the original ground branch is

$$e_{\text{one}}(\xi) = \frac{1}{4}b_2(-4\xi) - 1. \quad (\text{E17})$$

NIST DLMF 28.6.5 gives $b_2(q) = 4 - q^2/12 + 5q^4/13824 - 289q^6/79626240 + \dots$. The exact return in E17 is

$$e_{\text{one}} = -\xi^2/3 + 5\xi^4/216 - 289\xi^6/77760 + \dots,$$

matching the independent original character recurrence and the single-face row above. A modern original-source treatment of the single-plaquette Mathieu spectrum is Jakobs et al., EPJC85 (2025)1418, equation39; its Hamiltonian convention must be transported before numerical comparison. This literature check verifies the one-plaquette convention and coefficient. It does not claim that the full cubic-box E13 or the catalogue is new to the literature, nor supply an independent review of those results.

Sources: <https://dlmf.nist.gov/28.6.E5> ; <https://link.springer.com/article/10.1140/epjc/s10052-025-15120-x> . For the established exponential-vacuum and linked-source framework see Schütte, Zheng Weihong and Hamer, arXiv:hep-lat/9603026, section3.

F Explicit signed tangent certificate

This accompanies Q16 in `FOURTH_ORDER_SOURCE.md`. The source convention is the same original family of independent plaquette couplings, with `lambda_p=xi` only after the exact coefficients and their directional responses are retained.

The main coefficient producer solves the original eigenvector coefficient equation and then computes $Q_H \log u$. The independent program `checks/verify_signed_tangent.py` instead solves the logarithmic-source equation directly on every coefficient downset:

```
K v_nu = Q_H sum_(0<mu<nu) Gamma(v_mu,v_(nu-mu)),
K v_(e_p)=W_p.
```

Its returned order-four coefficient equals the entire stored catalogue polynomial for each of the 78 classes. The same polynomial ring, original link vector fields, and exact inverse residual are specified in Q5–Q13. The two coefficient constructions are distinct recurrences on those fixed objects.

For the original face p and each retained source multiindex ρ of total degree at most three, define

```
Z_(p,rho)=(rho_p+1) v_(rho+e_p).
```

The program evaluates the complete signed residual

```
K Z_(p,rho) - 2 Q_H sum_(0<mu<=rho)
      Gamma(v_mu,Z_(p,rho-mu))
```

and proves it is the zero quotient polynomial for positive degree ρ . At $\rho=0$ its value is exactly the original W_p . This is a finite formal-parameter coefficient identity, with every sign and multiplicity retained. It follows analytically by differentiating the logarithmic-source equation; the checker also executes all coefficient products explicitly.

The full receipt `results/signed_tangent.json` contains:

- 78 independent reconstructed logarithmic sources;
- 2,156 exactly zero polynomial tangent residuals;
- 282 explicitly expanded degree-three response polynomials, with the original differentiated face, remaining coupling multiindex, support multiset and chord coordinates for every one.

The original symmetry transports in `anchored_quartic_transports.json` return these representative polynomials to any original four-face cluster. The source-prefactor is the actual integer occurrence of that differentiated face; there is no extra factorial or loss of its original union label.

These are the signed finite coefficients through degree three in the derivative of the vacuum source. No new bound on the norm of an infinite linearized inverse, uniform gap or continuum limit is inferred from this finite coefficient data.